# THE USE OF HEDGES IN RESEARCH ABSTRACTS ACROSS APPLIED LIFE SCIENCES

A Thesis
Presented to the Faculty of
Department of English
College of Arts and Sciences
Pampanga State Agricultural University
Magalang, Pampanga

In Partial Fulfillment
of the Requirements for the Degree
BACHELOR OF ARTS IN ENGLISH LANGUAGE STUDIES

By

**CHRYZ KEAN F. MANUEL**

October 2026

# INTRODUCTION

## BACKGROUND OF THE STUDY

Certainty, it seems, is one thing academic writers are certain they should avoid. Writing guides urge developing researchers to "qualify your claims," as if hedging were a single transferable skill. Yet the actual language of uncertainty shifts with the nature of the evidence and the demands of each field.

Expert writers hedge to present claims with caution rather than absolute certainty. Hyland's work on scientific and molecular biology research writing shows that hedging is a core rhetorical resource: devices such as may, suggest, or generally signal epistemic stance, mark community membership, and let writers state claims with precision, caution, and deference to colleagues (Hyland, 1996, 1998). Those expert norms are well documented. Far less is known about undergraduate thesis writers, who face the same genre expectations without the same disciplinary training. Do their abstracts reflect field-specific conventions, or only a generic style of caution?

That question matters in the applied life sciences. Research in Agriculture, Veterinary Medicine, and Biology can shape community practice and environmental decisions. When developing researchers state provisional findings as settled fact, overstated claims can travel further than the evidence supports. Managing epistemic stance is therefore both a writing skill and a practical responsibility.

This study examines undergraduate thesis abstracts from three programs at Pampanga State Agricultural University (PSAU): Agriculture, Veterinary Medicine, and Biology. These fields share an institutional context

but differ in how they typically handle evidence. Agricultural research often involves field conditions outside the researcher's control. Biological studies frequently take place under controlled laboratory conditions. Veterinary medicine manages clinical variables tied to individual patient outcomes. The boundaries are not absolute. Agricultural work can be lab-based and biology can be field-based, but the different epistemic orientations may still shape hedging. If disciplinary norms appear at the undergraduate level, they should show up here. If they do not, the shared pattern is equally informative for instruction.

### *Hedging in Undergraduate Writing*

Developing researchers tend to underuse hedges and rely on a narrow set of devices, most often modal verbs (Khunasathitchai et al., 2025, Suratno & Aydawati, 2025, Ghia et al., 2022, and Mazidah et al., 2022). Across undergraduate and upper-level student writing, the same core pattern recurs: low hedge density, limited formal variety, and a default to modals such as may or local equivalents such as *dapat*, linked more to limited rhetorical exposure than to weak grammar. Thesis-abstract studies in Thai, Italian, and Indonesian L2 settings show averages near or below two hedges per abstract and, in some programs, heavier use of boosters than of hedges. Where disciplinary comparison is available among developing writers, as in Wang's (2022) sixteen-field MICUSP analysis, variation appears, yet modal dominance persists even in fields that hedge more overall.

These studies leave clear gaps for the present work. Only a few examined abstracts specifically, and most of those focused on English or language majors rather than applied science programs. Others analyzed discussion sections or

full papers, where rhetorical space is wider. None is set in a Philippine L2 context for Agriculture, Veterinary Medicine, and Biology undergraduate thesis abstracts.

*Disciplinary Variation in Scientific and Medical Writing*

Field norms shape expert hedging, but not along a simple hard/soft divide. Comparative work places humanities writers at the high end of hedge frequency and agriculture near the low end in one multi-field design (Afshar et al., 2014), while other pairings find scientists hedging more than applied linguists (Mohammed, 2011). Within life sciences, biology research articles often favor non-factive reporting verbs as the main hedge type (Souici & Slougui, 2022), and clinical writing may edge slightly above basic science within medicine without a strong statistical separation (Hamidi et al., 2025). The mixed directions matter less as a contradiction than as evidence that density tracks specific epistemic demands and coding choices rather than a single hard/soft divide.

These findings are not necessarily contradictory. Afshar et al. and Mohammed compare different disciplinary pairings and use different hedge taxonomies (Afshar the full Salager-Meyer (1994) scheme, Mohammed a lexical-hedge focus), so the contrast is evidence that hedging density tracks specific epistemic demands rather than a single hard/soft divide. Salager-Meyer (1994) frames hedging as claim-responsibility management: writers distribute responsibility for a claim's truth, accuracy, or applicability among themselves, their data, and external conditions. That framing predicts that Agriculture, Veterinary Medicine, and Biology may produce distinct patterns

not because one field is “harder,” but because each requires writers to manage different kinds of responsibility.

### *Hedging in L2 and EFL Contexts*

English is the medium of academic writing at PSAU, but not the home language for most students. L2 writers typically use fewer hedges and less formal variety than L1 peers (Demir, 2018, and Mohammed, 2011), and publication context can amplify that gap: English-medium journal abstracts show markedly more hedges than Chinese-medium counterparts in the same field (Hu & Cao, 2011). Explicit teaching can raise both frequency and variety among health-science undergraduates, though a preference for modals often remains even after training (Petchkij, 2019).

Because this study does not measure English proficiency independently, observed patterns may reflect L2 constraints, disciplinary socialization, or both. The aim is a descriptive baseline, not a causal account.

### *Hedging in Philippine Academic Writing Contexts*

Philippine and regional findings align with this picture. ASEAN learner data, including Philippine samples, show a preference for boosters over hedges and overuse of intensifiers such as completely, really, and always (Ningrum et al., 2024). Local college essay studies report developing-stage proficiency with epistemic modals dominant and only modest, often non-significant differences by major field (Escalona, 2025, and Lopez et al., 2025). Outside the student essay genre, Villanueva (2014) shows hedging used to manage accountability in Philippine newspaper editorials. Different genre, same point: hedging is context-sensitive rather than a fixed national style.

*Disciplinary Variation in Abstracts and Undergraduate Theses*

Abstracts compress research into a constrained space. Expert abstract studies report measurable disciplinary differences in linguistic features such as voice and nominalization (Imataki & Kobayashi, 2024), and large-scale work on promotion and caution finds hedge words more frequent in soft sciences and higher-ranking journals, with positive promotional wording more common in hard sciences (Xie & Mi, 2023). Expert abstract conventions are relatively well mapped. Developing-researcher abstracts are not. Whether undergraduates in applied life sciences show any disciplinary differentiation in this genre remains an open empirical question.

Across the literature, expert writers show measurable disciplinary differentiation in hedging shaped by epistemic demands rather than a simple hard/soft divide (Salager-Meyer, 1994, Afshar et al., 2014, Wang, 2022, and Xie & Mi, 2023). Developing researchers across contexts show limited hedge variety and tend to default to modal verbs (Khunasathitchai et al., 2025, Wang, 2022, Ghia et al., 2022, and Mazidah et al., 2022). L2 writers face additional constraints from linguistic factors, publication context, and limited explicit instruction (Demir, 2018, Hu & Cao, 2011, and Petchkij, 2019). Philippine and ASEAN studies confirm reliance on modals and, in some contexts, a preference for boosters (Ningrum et al., 2024, Escalona, 2025, and Lopez et al., 2025). Abstract-focused work further shows that caution and promotion vary by discipline and ranking (Xie & Mi, 2023), but undergraduate applied-science abstracts remain underdescribed.

No published study identified in this review has examined hedging in undergraduate thesis abstracts from applied science disciplines in a Philippine

L2 context. This study addresses that intersection at PSAU by mapping hedging patterns across three applied life sciences programs that share an institution but differ in their relationship to evidence.

## STATEMENT OF THE PROBLEM

Academic writing in the applied life sciences requires discipline-sensitive management of uncertainty. Prior work shows that expert hedging varies across fields such as agriculture and the basic sciences, yet there is no local corpus baseline for whether developing researchers at PSAU reflect those distinctions in their abstracts, or whether they rely on a generic, undifferentiated approach.

This study mapped hedging patterns in undergraduate thesis abstracts from Agriculture, Veterinary Medicine, and Biology at PSAU. Specifically, it asked the following questions:

1. What types of hedging devices appear in the undergraduate thesis abstracts across the three disciplines?
2. How does the frequency of hedging devices vary across Agriculture, Veterinary Medicine, and Biology programs?
3. Among the hedging devices identified, what rhetorical functions do they serve?
4. What do the frequency and functional patterns suggest about the construction of epistemic stance through hedging in the ratified abstracts of developing researchers at PSAU?

## RESEARCH FRAMEWORK

This study is grounded in a coding scheme adapted from Hyland's (1998) classification of hedging forms. Hyland (1998) is the instrument for this project. It supplies the grammatical types coded in the corpus (consolidated here as lexical hedges, modal verbs, adverbial hedges, and clause-level hedges) and the starting point for functional reading of retained hedges. Two function labels map onto that starting point. Epistemic Caution covers reduced commitment to the truth of a knowledge claim. Politeness/Deference covers interpersonal softening toward readers or prior work. Those two align roughly with content-oriented and reader-oriented concerns in the hedging literature. False positives are not a function label; they record candidates that fail both the Working Test and the Reader Test. The scheme is adapted for this corpus. It is not a fixed published taxonomy.

Each source used in the framework has a distinct job. Hyland (1998) is the coding instrument for hedge types and the broad start for function labels EC and P. Crompton (1997) is the boundary rule for Epistemic Caution alone. His restatement test (whether the claim could be restated with greater author commitment) operationalizes the Working Test for EC at Step 2. The study does not adopt Crompton's full taxonomy or his narrower view of what may enter the retained set. Salager-Meyer (1994) is an interpretive lens used in Part IV. Her claim-responsibility account helps read why density may differ across programs. Her taxonomy is not used for coding. Hyland (2005) is a second interpretive lens. The stance model helps read what fewer hedges mean where results turn into implications. It is not an operational coding criterion.

Together, the two lenses frame hedging as both an epistemic and an interpersonal resource without treating either lens as a tested finding.

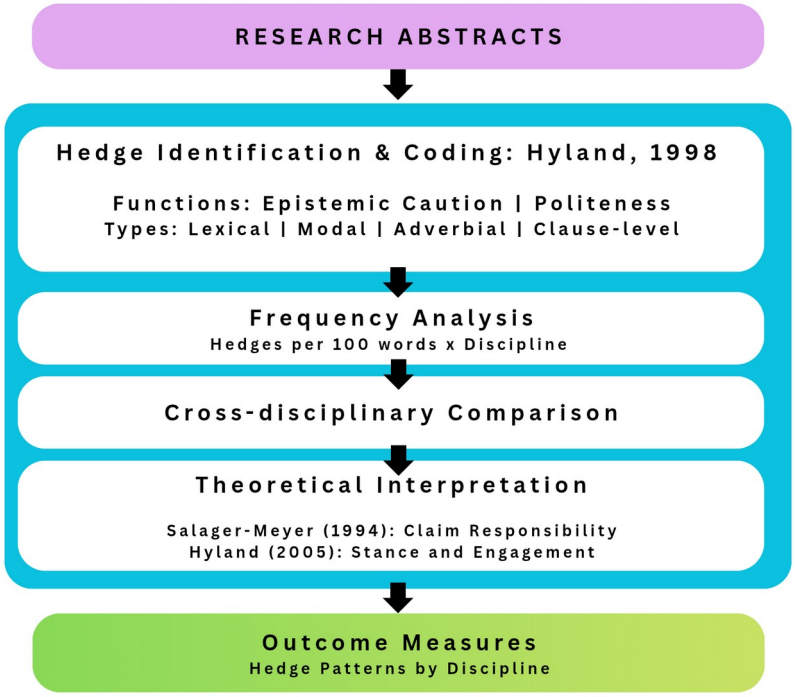
In practice, the coding scheme was adapted from Hyland (1998) as the type instrument and operationalized through the decision rules in Appendix A. After a holistic reading of each abstract, coding followed three stages. Step 1 marked all possible hedge candidates (wide net). This step did not yet count retained hedges. Step 2 identified function with two paired removal checks. The Working Test asks whether removing the device changes commitment to the truth of the proposition. If it does, the token is retained as Epistemic Caution (EC). The Reader Test asks whether, when truth-commitment stays the same, removing the device makes the claim more imposing, unattributed, or confrontational toward readers or prior work. If it does, the token is retained as Politeness/Deference (P). If neither check is met, the token is not a hedge and is logged as a false positive (FP). FP is a residual record, not a function label. The only function labels are P and EC (EC+P when both checks hold on one token). Step 3 assigned grammatical type (modal, lexical, adverbial, clause-level) to retained P and EC tokens only. In this corpus all 65 retained hedges were EC. No P forms appeared. RQ3 reports the functional profile of retained hedges (P versus EC). RQ4 is answered from frequency patterns together with interpretive reading under the lenses named above. Figure 1 presents the study overview: research abstracts through Hyland-based identification and coding (functions EC and P; types lexical, modal, adverbial, clause-level), frequency analysis, cross-disciplinary comparison, and theoretical interpretation, yielding hedge patterns by discipline.

The only function labels in this study are Epistemic Caution (EC) and Politeness/Deference (P). False positives are not a function. They are a residual record of candidates that meet neither removal check. EC and P may co-occur on one retained token (EC+P) when both the Working Test and the Reader Test hold. Connectors and framing forms may appear in the Step 1 net. If neither test is met, they are recorded as FP, not as a third function. It is not a license to count every soft-sounding word. Crompton (1997) informs only the EC diagnostic. A proposition restatable with greater author commitment signals reduced epistemic weight. Crompton argues that the term hedge is most useful when reserved for such epistemic provisionality on knowledge claims, rather than extended to politeness-related features such as passives, impersonal constructions, or emotion-projecting lexis as such. The present study does not adopt Crompton's full taxonomy. It uses that argument to support the EC diagnostic on retained hedges, not as the sole candidate-tagging rule. The 100% EC profile of the retained set is therefore readable as both an empirical pattern in this sample and a product of the Step 2 rule that keeps only claim mitigation in the hedge tally. The zero for P is the clearer empirical null in this abstract genre. Non-qualifying candidates are FP records, not a third function.

The conceptual framework is presented in Figure 1. The grouping variable is discipline (Agriculture, Veterinary Medicine, and Biology). The outcome measures are hedge type frequency, distribution, and function. Figure 1 shows the study's type and function categories in that pipeline. The analytical path moves from Step 1 and Step 2 through frequency analysis and cross-

disciplinary comparison to functional and interpretive reading. Source jobs are stated in the framework text: Hyland (1998) supplies the type instrument and the starting point for EC and P on retained hedges, Crompton (1997) supports the Working Test for EC, Salager-Meyer (1994) is the Part IV claim-responsibility lens, and Hyland (2005) is the stance lens for implications. Figure 1 shows functions as Epistemic Caution and Politeness only; false positives are a residual coding record, not a function category in the figure. Two boundary conditions are acknowledged throughout. The texts are produced by developing researchers whose work may not yet reflect discipline-specific norms. The institutional context at PSAU includes L2 English use, but language proficiency is not measured as a variable in this design. These factors bound how far the findings may be generalized.

**Figure 1**




*Conceptual Framework of the Study*

## SIGNIFICANCE OF THE STUDY

**For writing instructors and thesis advisers at PSAU,** the study provides an empirical baseline of how uncertainty is managed in the ratified abstracts of three applied life sciences programs. That baseline can support more targeted feedback than generic advice to "be cautious."

**For academic writing researchers,** it adds Philippine applied-science abstract data to a body of work that has drawn mainly from other regional and disciplinary settings.

**For the BAELS program,** it shows that corpus-assisted discourse analysis is a viable undergraduate thesis methodology and yields findings usable in local writing pedagogy.

**For future researchers,** the sampling design, coding decision rules, and false-positive log offer a replicable protocol for similar corpus studies.

## SCOPE AND LIMITATIONS

This study focused on undergraduate thesis abstracts from Agriculture, Veterinary Medicine, and Biology at PSAU. The inclusion window refers to the year of hard-bound submission as recorded in the PSAU library catalog, not the year the research was conducted. Theses conducted earlier but bound and catalogued in 2023–2025 are included. This reflects institutional practice in which submission date, particularly for irregular students, may postdate the research period by several years. The corpus consisted of 60 abstracts (20 per discipline, with Agriculture split evenly between Crop Science and Animal Science). Sampling design, stratum intervals, and inclusion/exclusion criteria are detailed under Research Corpus. Table 1 reports population and sample sizes. Sub-programs are analyzed as a single “Agriculture” discipline for cross-disciplinary comparison. No separate inferential comparison between them is undertaken.

For a manually coded functional analysis within a single institution, 60 texts are treated as sufficient for identifying patterns within the limits of a BA thesis. Findings are indicative patterns in the PSAU context. Exploratory inferential analyses (chi-square, Kruskal-Wallis) are reported transparently and should not be treated as confirmatory given the small sample and zero inflation. Comparable undergraduate corpus sizes and the larger multi-field

MICUSP work are noted under Research Corpus. The pilot comparison appears in Appendix B.

The study is delimited to complete English-language abstracts. Abstracts with significant Filipino code-switching are excluded to maintain linguistic consistency. Disciplinary classification followed the degree program under which each thesis was officially submitted, not a research topic, to ensure objective categorization. L2 proficiency is not measured as a variable. Where patterns suggest proficiency effects alongside disciplinary ones, this is noted as a limitation requiring future research with proficiency controls.

The abstract genre's rhetorical compression may suppress hedging variety compared to full-text discussions. Findings reflect hedging in this constrained genre only and should not be extrapolated to developing researchers' hedging competence in longer texts without further research. Additionally, this study is primarily delimited to hedging devices. A limited count of assertive reporting forms was performed only on the 25 zero-hedge abstracts. No comparison with hedged abstracts was made, and a full booster analysis was not undertaken. That count is descriptive of the zero-hedge subset only. While credibility in academic writing involves a balance of caution and certainty, this study focuses specifically on the management of uncertainty, which is often the more challenging rhetorical skill for developing L2 researchers to acquire.

Reliabili

ty was assessed through intra-rater agreement only. A second coder was not available for inter-rater reliability testing, which limits how far the coding scheme can be verified against an independent application of the same decision

rules. That limit bears especially on the functional findings in Part III. Assigning all 65 retained hedges to Epistemic Caution may reflect one coder's frame as much as a property of the corpus. Inter-rater coding on at least a subset of the corpus is needed before those functional results can be treated as established. Related work could apply the same Step 1–3 coding scheme to longer genres where P may appear more often and where framing-only candidates can be compared across genres.

The study also lacks a comparable expert-abstract corpus against which to compare the functional findings. Among retained hedges, no tokens were independently labeled Politeness (P = 0). False positives (n = 58) record non-qualifying candidates from the same pool. That result describes the retained set in this sample, not the absence of interpersonal mitigation from student writing overall. Genre compression in abstracts is a plausible account of why P items did not appear in the retained set. Expert abstracts in the same three disciplines, coded with the same Step 1–3 coding scheme, would help separate genre effects from writer-competence effects.

Crucially, this study analyzes the archived thesis abstract as a ratified textual product rather than a measure of individual cognition. Undergraduate theses are heavily supervised, and abstracts are often the product of multiple revisions guided by advisers. Consequently, observed hedging patterns do not solely reflect the independent rhetorical choices or internalized norms of developing researchers. Rather, they reflect the standards of the disciplinary community as enforced by faculty advisers. Therefore, while the study investigates patterns within texts produced by developing researchers, its

claims are limited to the epistemic stance constructed in the final abstracts and the departmental standards they represent, rather than the internal cognitive processes of the writers themselves. The findings are strictly descriptive of the observed patterns within the PSAU context.

## DEFINITION OF TERMS

The following terms are defined conceptually and operationally to ensure consistency throughout the study:

**Abstract,** conceptually, is a concise, self-contained summary of a research work. Operationally, this refers to the complete English-language summary section of undergraduate theses archived in the PSAU institutional repository from 2023 to 2025. It contains research objectives, methods, results, and conclusions.

**Adverbial Hedges,** conceptually, are adverbs or adverbial expressions that qualify the scope, frequency, or certainty of a proposition. Operationally, these include generally, typically, possibly, probably, relatively, and essentially. These are coded in this study when they restrict the generalizability of a claim rather than describe a procedural norm.

**Applied Life Sciences,** conceptually, refer to disciplines that apply scientific knowledge to the study and management of living organisms and biological systems. Operationally, this refers to the three specific undergraduate programs at PSAU included in the study: Agriculture, Biology, and Veterinary Medicine.

**Claim Responsibility,** conceptually, refers to the epistemic obligation a writer accepts for the accuracy, truth, or limitations of a statement (Salager-Meyer, 1994). Writers use hedges to distribute this responsibility between

themselves, their data, or external factors. Operationally, this serves as the interpretive lens for analyzing why specific hedges were chosen in the abstracts.

**Chi-Square Test of Independence,** conceptually, evaluates whether two categorical variables are associated. Operationally, this study used it to test the association between discipline and zero-hedge status (zero vs. nonzero). The test statistic is $\chi^2 = \Sigma [(O_i - E_i)^2 / E_i]$, where $O_i$ is the observed frequency in cell i and $E_i$ is the expected frequency under independence. Degrees of freedom = (rows − 1) × (columns − 1). The reported result was $\chi^2 = 5.07$, df = 2, p = .079.

**Clause-Level Hedges,** conceptually, are syntactic structures that frame a proposition as tentative, approximate, or subject to qualification. Operationally, these include expressions such as it seems that, it is possible that, to some extent, and in our view. These are coded in this study when the entire clause functions to express the writer's epistemic reservation about the following or embedded proposition.

**Cohen's Kappa (κ),** conceptually, measures inter- or intra-rater agreement for categorical coding beyond chance. Operationally, this study used it for intra-rater reliability on a second pass of six abstracts. The formula is $\kappa = (P_o - P_e) / (1 - P_e)$, where $P_o$ is the observed proportion of agreement and $P_e$ is the expected agreement by chance. Landis and Koch (1977) interpret κ values as: <0.00 poor, 0.00–0.20 slight, 0.21–0.40 fair, 0.41–0.60 moderate, 0.61–0.80 substantial, 0.81–1.00 almost perfect. The intra-rater result was $\kappa = 0.89$.

**Corpus-Assisted Discourse Analysis,** conceptually, uses systematic text collections to identify linguistic patterns, combining quantitative frequency measures with qualitative functional interpretation. Operationally, this refers

to the manual frequency counting, functional coding, and cross-disciplinary comparison of hedging devices across 60 PSAU thesis abstracts using a Google Sheets coding template.

**Disciplinary Variation,** conceptually, refers to systematic differences in language use across academic fields arising from different epistemic norms, evidence types, and rhetorical conventions. Operationally, this refers to observable differences in hedging frequency, type, and function across Agriculture, Veterinary Medicine, and Biology abstracts in this corpus.

**Epistemic Stance,** conceptually, refers to the linguistic expression of a writer's degree of commitment to the truth, certainty, or reliability of a proposition. It signals the writer's judgment about the status of the knowledge they are presenting. Operationally, this refers to the aggregate pattern of hedging devices used in an abstract. A high frequency of hedges indicates a cautious or tentative stance, while a low frequency or absence of hedges records fewer overt markers of caution. It does not by itself establish absolute commitment, especially where findings are binary or measured.

**Epistemic Weight, conceptually, is the strength of the writer's commitment to the truth of a proposition (how certain or tentative the claim is). Operationally, a change in epistemic weight supports an EC label under the Working Test at Step 2. The Reader Test at Step 2 tracks interpersonal softening (imposition, attribution, confrontation toward readers or prior work) for a P label. Neither check met yields FP, not a function.**

**Hedge Function, conceptually, refers to the rhetorical purpose a hedging device serves in its discourse context. Operationally, this study uses two function labels only, adapted from Hyland's (1998) discussion of hedging**

**functions (not a fixed published taxonomy): Epistemic Caution (EC), which expresses reduced commitment to a claim's truth or accuracy, and Politeness/Deference (P), which shows interpersonal softening toward readers or prior research. False positives are a residual coding record, not a function.**

**Hedging,** conceptually, is the use of linguistic devices to express caution or tentativeness in academic writing, serving as a mechanism for managing claim responsibility (Salager-Meyer, 1994). Operationally, this refers to four device categories identified in the corpus following Hyland's (1998) taxonomy. These categories are lexical hedges, modal verbs, adverbial hedges, and clause-level hedges.

**Kruskal-Wallis Test,** conceptually, is a non-parametric rank-based alternative to one-way ANOVA for comparing distributions across three or more independent groups. Operationally, this study applied it to per-abstract hedge rates (hedges per 100 words) across the three disciplines. The test statistic is $H = [12 / (N(N+1))] \times \Sigma (R_j^2 / n_j) - 3(N+1)$, where N is the total number of observations, $n_j$ is the size of group j, and $R_j$ is the sum of ranks for group j. The reported result was $H = 7.44$, $p = .024$.

**Lexical Hedges,** conceptually, are verbs, adjectives, or nouns that introduce uncertainty or tentativeness at the propositional level. Operationally, these include verbs such as suggest, indicate, appear, and seem. They also include adjectives such as possible, probable, and likely, and nouns such as possibility and assumption. These are coded in this study when they express epistemic qualification of a claim rather than description or recommendation.

**Mann-Whitney U Test,** conceptually, is a non-parametric test that compares the distributions of a continuous or ordinal variable between two independent groups. Operationally, this study used it as a post hoc follow-up after the Kruskal-Wallis test, comparing per-abstract hedge rates (hedges per 100 words) for each pair of disciplines. The test statistic is $U = n_1n_2 + [n_1(n_1+1)]/2 - R_1$, where $n_1$ and $n_2$ are the sample sizes of the two groups and $R_1$ is the sum of ranks for group 1. A Bonferroni correction for three pairwise tests set $\alpha = .017$. The reported results were Veterinary Medicine versus Biology $p = .011$, Veterinary Medicine versus Agriculture $p = .034$, and Agriculture versus Biology $p = .72$. Only the Biology–Veterinary Medicine contrast remained significant after correction.

**Modal Verbs,** conceptually, are auxiliary verbs that express degrees of possibility, probability, or conditionality. Operationally, these include may, might, could, would, should, and can. These are coded in this study when they signal epistemic uncertainty about a claim rather than capability, obligation, or future procedure.

**Normalized Frequency,** conceptually, is a statistical method used to compare linguistic features across texts of different lengths by standardizing the count to a base unit. Operationally, this refers to the formula used in this study to compare hedge usage: Normalized Frequency = (Total Hedges / Total Word Count) x 100. For example, if an abstract contains 3 hedges and 258 words, the normalized frequency is (3/258) x 100 = 1.16 hedges per 100 words.

**Operational Restatement (Threshold Restatement),** conceptually, a phrase that only restates a result already fixed by the measure or a field-standard threshold, without a further uncertain inference. Operationally,

evidential verbs used this way (especially indicating) are coded FP-Gen rather than as lexical hedges. In Results this rule is also called the “indicating” tautology test.

**Rescue-Hedging (Rescue Framing),** conceptually, the use of hedges after an explicit non-significant result so that a treatment or finding is still presented as potentially beneficial. Operationally, the term is used in this study as a descriptive label for that pattern.

**Softening, conceptually, is the reduction of force, directness, imposition, or interpersonal impact in a statement. Operationally, Step 2 tracks softening of force or directness on knowledge claims (and interpersonal imposition on how claims are put forward). Retained hedges are labeled EC and/or P only, by the Working Test and Reader Test at Step 2. FP is not a function label. Directives remain excluded (FP-Rec).**

# METHODOLOGY

This chapter describes the research design, institutional context, data collection process, and corpus selection for the study. Each methodological decision was justified by the nature of the research questions and the practical constraints of an undergraduate thesis.

## RESEARCH CONTEXT

This study was situated at Pampanga State Agricultural University (PSAU), a state university specializing in agricultural and applied life sciences. Agriculture, Veterinary Medicine, and Biology are the core applied life science programs of the university, making them the most appropriate disciplines for an institutional corpus study.

The Philippine higher education context is relevant because English is the official medium of academic writing instruction at PSAU despite being a second language for most of the population. At PSAU, undergraduate writing instruction in the programs sampled for this study emphasizes structure and grammatical correctness more than stance. Hedging is not treated as a distinct rhetorical skill in the course materials reviewed for this project. Developing researchers writing thesis abstracts were therefore producing a genre that demanded hedging with limited explicit preparation for its epistemic requirements. These conditions made PSAU an appropriate institutional site for examining disciplinary hedging among developing researchers in a Philippine L2 context.

The PSAU thesis guidelines available for this study prescribed standard formatting but did not set a rigid rhetorical template for abstracts.

## RESEARCH DESIGN

This study employed a descriptive mixed-methods design combining quantitative frequency analysis with qualitative functional interpretation (Biber et al., 1998). The quantitative layer identified hedging type and distribution patterns through frequency counts normalized per 100 words. This enabled cross-disciplinary comparison on a common scale. The qualitative layer involved functional coding using Hyland's (1998) framework to assign rhetorical purposes to the identified hedges. The interpretation of disciplinary patterns was then informed by two theoretical lenses. These are Salager-Meyer's (1994) concept of claim-responsibility and Hyland's (2005) model of stance and engagement. All coding was conducted manually using Google Sheets. This prioritized researcher judgment over automated processing for functional coding decisions, since automated tools cannot reliably assign hedge function from context, which was central to this study's research questions.

## RESEARCH PROCESS

The study proceeded through the following steps:

**a. Requesting Permission.** The researcher secured written permission from the PSAU Library administration to access the physical archives of undergraduate theses. Due to institutional policies on intellectual property, soft copies of theses are restricted and cannot be provided electronically. Access is limited to on-site reading of hard-bound copies. The researcher obtained

specific authorization to photograph the abstract sections of the selected theses for research purposes.

**b. Corpus Construction.** Disproportionate stratified systematic sampling was used, with discipline as the stratification variable and fixed analytical targets per stratum to ensure equal representation regardless of population size. The target sample sizes were n=20 for Biology, n=20 for Veterinary Medicine, n=10 for Agriculture, Animal Science, and n=10 for Agriculture, Crop Science, totaling 60 abstracts. Within each stratum, a sampling interval k was calculated as k = N/n, where N is the stratum's target pool size (theses completed 2023–2025) and n is the required sample size. The intervals were as follows: Biology k=7 (145÷20=7.25), Veterinary Medicine k=5 (104÷20=5.2), Animal Science k=4 (38÷10=3.8), and Crop Science k=9 (94÷10=9.4). Sampling began from a randomly selected position around the middle of each stratum's catalog and proceeded forward by taking every kth thesis. Where the end of the stratum was reached before the required sample size, counting wrapped around to the beginning of the catalog and continued until the target sample was met. Disciplinary classification followed the degree program under which each thesis was officially submitted (e.g., B.S. Agriculture, B.S. Biology, D.V.M.), not a research topic, to ensure objective categorization.

**c. Text Extraction.** Abstracts were captured through on-site photography of the hard-bound thesis copies. These images were then converted to plain text using Optical Character Recognition (OCR) tools such as Google Drive or Adobe Scan. Each converted text was manually verified against the original photographs to correct transcription errors. Particular attention

was paid to verifying sentences containing potential hedge tokens, as OCR errors in these sentences would directly affect the coding results. Text files were anonymized using discipline codes (Acs1–Acs10, Aas1–Aas10, VM1–VM20, B1–B20) and organized by discipline.

**d. Pilot Testing and Reliability Checking.** A pilot test was conducted on 4 abstracts not included in the final corpus to establish the overall coding scheme (see Appendix A). Decision rules for boundary cases were refined during main-corpus coding as specific ambiguities appeared and logged in the codebook. After the codebook was finalized, the full corpus was re-coded under the final rule set. The intra-rater reliability check (second pass on 6 abstracts) used that same final codebook. Pilot abstracts were drawn from the PSAU library archives but from theses whose hard-bound submission year was before 2023, keeping them outside the study's inclusion window. To ensure reliability, the study employed a two-step verification process. First, intra-rater consistency was checked by double-coding approximately 10% of the corpus (6 abstracts, 2 per discipline) after a one-week interval, yielding 95% exact agreement (Cohen's $\kappa = 0.89$, indicating almost perfect agreement per Landis & Koch, 1977). Second, the coded data were reviewed by the thesis adviser to verify consistency and accuracy. This approach combined intra-rater reliability with adviser-supervised validation, which was the standard practice for single-researcher undergraduate projects. Discrepancies were resolved through discussion and consensus.

**e. Data Analysis.** Hedge identification, type classification, and functional coding were all conducted manually using a Google Sheets coding template. Each abstract was read in full before coding began to ensure hedges were

interpreted in context rather than flagged as isolated tokens. Ambiguous cases were resolved using the predefined decision tree in Appendix A. Frequency counts were normalized per 100 words to enable comparison across abstracts of varying length. In a separate pass, each abstract was also coded for outcome type from its results language (Binary, Continuous, Mixed, Null/NS, Other), so that density patterns could be read against the kind of finding reported. Those codes are recorded in the coding spreadsheet and discussed in Part IV. While the quantitative analysis identified what hedges were used, the qualitative interpretation applied Salager-Meyer's (1994) claim-responsibility framework and Hyland's (2005) stance model to explain why these patterns occurred. Given the corpus size, findings were interpreted as indicative patterns within the PSAU context, supplemented by inferential analyses (chi-square, Kruskal-Wallis) whose results are reported transparently but should not be treated as confirmatory given the small, zero-inflated sample.

**f. Timeline.** Permission and library access: 1 week. Corpus construction and on-site photography: 3 weeks. OCR verification and text preparation: 2 weeks. Pilot testing and coding finalization: 2 weeks. Full coding and frequency analysis: 8 weeks (approximately 10 hours per week). Writing and revision: 8 weeks. Total timeframe: approximately 5.5 months (24 weeks).

## CODING DECISION RULES FOR AMBIGUOUS CASES

Ambiguous forms such as can, potential, indicating, and possibility may function either as epistemic hedges or as non-epistemic lexical items depending on context. The following decision rules govern these cases. First, "can" combined with an active content verb describing a treatment effect (enhance, promote, reduce, help) is classified as capability (FP-Cap) regardless

of whether it appears inside a “suggests that” clause, because the Working Test shows that removing “can” alone does not change the epistemic weight of the clause when “suggests” is already carrying the hedge. Second, the “indicating” tautology test covers not only numeric restatement but also field-standard interpretive thresholds: “non-toxic” from no mortality observed, “limited benefits” from poor performance, and “superior feed efficiency” from lower feed conversion ratio are treated as operational restatements (FP-Gen) rather than inferential hedges. Third, for “potential + noun” constructions, structural position is the primary disambiguator. Aim, methods, and pure topic-naming frames are coded FP-Gen, as when an abstract “explores the potential antifungal properties” of a plant or names “potential pharmaceutical applications” as the object of further study. Results and conclusion frames that qualify a finding-claim are coded Lexical EC. The Working Test applies as a secondary check. When removing the token clearly changes commitment to a proposition, for example when “presence of potential mycobacterial infections” becomes an asserted presence claim without the modifier, the token is coded Lexical EC even in an aim or rationale sentence. The same position-based treatment applies to “possibility + noun” in aim and method sentences. Fourth, when a sentence contains multiple hedge-like tokens, each token is tested individually in isolation, ignoring the other hedges in the sentence. A token counts only if removing it alone changes the epistemic weight of its own clause. The corpus therefore reflects contextual rather than purely lexical classification, and the 58 rejected candidates document the decision trail for tokens that were considered but excluded. Additionally, the outlier abstract Acs2 uses a repeated hedging template (“This suggests that while X might not

have been Y, Z could still [verb] beneficial effects") twice with light variation, inflating its own token count through formulaic repetition rather than independent hedging decisions.

## RESEARCH CORPUS

This study used a text corpus rather than human participants. The corpus consisted of 60 undergraduate thesis abstracts from Agriculture, Veterinary Medicine, and Biology programs at PSAU. This study employed disproportionate stratified systematic sampling, with discipline as the stratification variable and fixed analytical targets per stratum (n=20 for Biology, n=20 for Veterinary Medicine, n=10 for Animal Science, and n=10 for Crop Science) to ensure equal representation regardless of population size. This meant the sampling rate varied by population size, prioritizing equal sample sizes over proportional representation. For the Agriculture discipline, the sample was equally split between Crop Science (n=10) and Animal Science (n=10) to ensure balanced representation of the sub-disciplines. Within each stratum, a sampling interval k was calculated as $k = N/n$. The intervals were Biology k=7 (145÷20=7.25), Veterinary Medicine k=5 (104÷20=5.2), Animal Science k=4 (38÷10=3.8), and Crop Science k=9 (94÷10=9.4). Sampling began from a randomly selected position around the middle of each stratum's catalog, proceeded forward by taking every kth thesis, and wrapped around to the beginning of the catalog when needed until the target sample was met. This sample size is appropriate for the study's scope. Khunasathitchai et al. (2025) used 60 abstracts in a comparable corpus study. Wang (2022) examined a much larger multi-field student corpus. The present study does not claim a comparable scale. For a manually coded functional analysis at a single

institution, 60 texts allow sufficient depth of interpretation within the timeframe and resources of a BA thesis. The study treated findings as baseline descriptive data for institutional curriculum review.

**Table 1**

PSAU Thesis Repository: Population and Sampling Summary

| Category | Biology | Veterinary Medicine | Animal Science | Crop Science | Total |
|---|---|---|---|---|---|
| Total repository | 275 | 206 | 131 | 216 | 828 |
| 2023–2025 (target pool) | 145 | 104 | 38 | 94 | 381 |
| Pre-2023 (excluded pool) | 130 | 102 | 93 | 122 | 447 |
| Pilot sample (pre-2023) | 1 | 1 | 1 | 1 | 4 |
| Main sample | 20 | 20 | 10 | 10 | 60 |

Table 1 presents the repository populations and the final sample distribution. The target pool for 2023–2025 totals 381 theses. The equal allocation strategy ensures that each discipline provides a robust sample size for cross-disciplinary comparison, preventing the smaller Animal Science population from being underrepresented in the analysis. Pre-2023 theses (n=447) serve as the source pool for pilot abstracts to keep them distinct from the main study period.

**Inclusion Criteria.** English-language undergraduate thesis abstracts. Complete abstracts available in the PSAU library archives. Clear disciplinary classification based on official degree program. Hard-bound submission year (as recorded in the PSAU library catalog) between 2023 and 2025.

**Exclusion Criteria.** Non-English abstracts or abstracts with significant Filipino code-switching. Graduate-level theses. Theses submitted under interdisciplinary degree programs (e.g., Environmental Science) where a single disciplinary affiliation cannot be determined. Theses whose hard-bound submission year (library catalog) is before 2023.

**Ethical Considerations.** Abstracts were accessed on-site at the PSAU Library with proper authorization. Student names were fully anonymized using discipline codes (Acs1–Acs10, Aas1–Aas10, VM1–VM20, B1–B20) before analysis began. Thesis titles that appear as quoted wording inside an abstract (source text) were redacted in the coded appendix where they identified the work. Residual title fragments, if any, do not include student names. The study analyzed linguistic patterns in text, not student performance or personal information. All data files were stored on password-protected devices and encrypted cloud storage accessible only to the researcher and thesis adviser. These files are retained for five years to support verification of coding decisions by future researchers and to allow for follow-up comparative studies, after which they are permanently deleted in line with standard academic data retention practice.

# RESULTS, ANALYSIS, AND DISCUSSION

This chapter presents the results, analysis, and discussion of the hedging patterns identified in the 60 undergraduate thesis abstracts across Agriculture, Veterinary Medicine, and Biology at Pampanga State Agricultural University. The chapter is organized into four parts, each corresponding to a research question. Part I presents the types of hedging devices found across the three disciplines. Part II examines how hedge frequency varies across Agriculture, Veterinary Medicine, and Biology. Part III identifies the rhetorical functions served by the hedging devices. Part IV discusses the main cross-disciplinary contrast in the ratified abstracts of developing researchers at PSAU.

## PART I: HEDGE TYPES ACROSS THE THREE DISCIPLINES

The coding identified 65 hedge tokens across the 60-abstract corpus, or about 1.08 hedges per abstract. That rate sits near the lower end of published abstract figures. Hu and Cao (2011) reported about 2.04 hedges per abstract in English-medium applied linguistics journals and lower rates in Chinese-medium journals. These tokens were classified into four types following Hyland's (1998) taxonomy: modal verbs, lexical hedges, adverbial hedges, and clause-level hedges. Table 2 presents the distribution of hedge types across the full corpus.

**Table 2**

Hedge Type Distribution Across the Corpus

| Hedge Type | Count | Percentage |
|---|---|---|
| Modal Verbs | 21 | 32.3% |
| Lexical Hedges | 41 | 63.1% |
| Adverbial Hedges | 3 | 4.6% |
| Clause-Level Hedges | 0 | 0.0% |
| Total | 65 | 100.0% |

Table 2 presents the overall distribution of hedge types. Lexical hedges dominated the corpus with 41 instances (63.1%), followed by modal verbs with 21 instances (32.3%), and adverbial hedges with 3 instances (4.6%). No clause-level hedges were identified in any of the 60 abstracts. This finding partially confirms the pattern reported in the literature, where modal verbs are typically the most frequent hedge type in developing researchers′ writing (Khunasathitchai et al., 2025, and Wang, 2022). In this corpus, however, lexical hedges outnumber modal verbs, a pattern that diverges from the modal-dominated writing documented in prior studies.

Within the lexical hedge category, the token “potential” accounted for 19 of the 41 lexical instances (46.3%), the single most frequent hedge token in the corpus. Other frequent lexical forms, grouped by root, included “suggest / suggests / suggesting” (13), “imply / implies / implying” (3), and “indicating” (2). Modal verbs were led by “can” (6), “could” (6), “may” (6), and “might” (3). Concordance of the 19 potential instances shows recurring frames such as “the potential of X as a Y,” “has potential as an X,” and “highlight their potential as X.” That recurrence is notable. Whether those uses express epistemic caution or topic-naming evaluation remains a boundary judgment under the Working Test and is open to alternative readings. Agriculture and Biology account for 15 of the 19 instances. Veterinary Medicine accounts for 4. Table 3 lists each

retained form so the lexical subtotal of 41 and the corpus total of 65 can be checked form by form.

**Table 3**

Hedge Token Inventory by Form

| Hedge form | Grammatical type | Count | % of total hedges |
|---|---|---|---|
| potential | Lexical Hedges | 19 | 29.2% |
| suggests | Lexical Hedges | 7 | 10.8% |
| suggest | Lexical Hedges | 3 | 4.6% |
| suggesting | Lexical Hedges | 3 | 4.6% |
| implies | Lexical Hedges | 2 | 3.1% |
| indicating | Lexical Hedges | 2 | 3.1% |
| likely | Lexical Hedges | 2 | 3.1% |
| implying | Lexical Hedges | 1 | 1.5% |
| inferring | Lexical Hedges | 1 | 1.5% |
| possible | Lexical Hedges | 1 | 1.5% |
| can | Modal Verbs | 6 | 9.2% |
| could | Modal Verbs | 6 | 9.2% |
| may | Modal Verbs | 6 | 9.2% |
| might | Modal Verbs | 3 | 4.6% |
| potentially | Adverbial Hedges | 3 | 4.6% |
| Total | | 65 | 100.0% |

Table 3 lists every retained hedge form in the corpus with its grammatical type and count. Counts sum to 65 and match Table 2. Surface forms of the same root are listed separately here so each tally line can be checked against the coding sheet.

The distribution of hedge types shows cross-disciplinary differences, most clearly in Veterinary Medicine's small total and low modal share. Table 4 presents the per-discipline breakdown of hedge types.

**Table 4**

Hedge Type Distribution by Discipline

| Discipline | Modal Verbs | Lexical Hedges | Adverbial | Clause-Level | Total |
|---|---|---|---|---|---|
| Agriculture | 14 (51.9%) | 13 (48.1%) | 0 | 0 | 27 |
| Biology | 5 (17.9%) | 20 (71.4%) | 3 (10.7%) | 0 | 28 |
| Veterinary Medicine | 2 (20.0%) | 8 (80.0%) | 0 | 0 | 10 |
| Total | 21 (32.3%) | 41 (63.1%) | 3 (4.6%) | 0 | 65 |

Table 4 shows that Agriculture has a nearly balanced type mix (14 modal verbs, 51.9% vs. 13 lexical hedges, 48.1%), while Biology is heavily lexical (20 of 28 hedges, 71.4%, with only 5 modal verbs). Veterinary Medicine, with only 10 total hedges, is also predominantly lexical (8 of 10, 80.0%). Agriculture is the only discipline in which modal verbs approach half of the hedge tokens. Biology and Veterinary Medicine lean lexical. That contrast is real in the counts, but the Ag–Bio difference in overall density is marginal and outlier-sensitive (Part II), so type-mix differences should not be read as a stable Agriculture-versus-Biology profile. The near-absence of adverbial hedges across all three disciplines (only 3 instances, all in Biology) points instead to a shared, narrow repertoire.

The absence of clause-level hedges is consistent with the compressed abstract genre and with the supervised character of the corpus. Science abstracts favor impersonal, compact reporting. Clause-level frames that foreground the writer's stance (including first-person forms such as "I think" or "in our view," and more impersonal ones such as "it seems that") are uncommon in that register. The Scope and Limitations section notes that the texts are ratified products shaped by adviser feedback, so the zero count is better read as a feature of the genre and of departmental practice than as evidence that these writers lack the syntactic means to hedge at clause level.

The same restricted range of devices has been observed in other developing-writer corpora (Hyland & Milton, 1997). Here the institutional and genre setting supplies the more proximate explanation.

Alongside the 65 retained EC hedges (and P = 0), the corpus yielded 58 false positive tokens—hedge-like words rejected under Step 2 (Working Test + Reader Test). Table 6 places these totals side by side. The false positive breakdown shows FP-Gen (general exclusion, including topic-naming and evidential pointers) at 33 instances, FP-Rec (recommendations) at 15, FP-Proc (procedural norms) at 2, and FP-Cap (capability) at 8. The high number of FP-Gen instances, particularly involving the word “potential” in aim and methods statements, reflects a systematic ambiguity in the corpus: the same word functions as a hedge in conclusion sentences but as topic-naming in introduction and methods sentences. This distinction was codified in the codebook’s rule for “potential + noun” constructions, with structural position as the primary disambiguator and the Working Test as a secondary screen. Without that position-based rule, keyword matching alone would have counted many topic-naming uses of potential as hedges and inflated the total.

Additionally, 14 unhedged assertions were identified where caution was warranted but absent. Of these, 9 were generalizations beyond the data [G] and 5 were causal claims stated more strongly than the design alone supports [C]. Twelve of the 14 fall in zero-hedge abstracts. Bare trial results and measured findings are not counted: a hedge could often be inserted there without changing the facts, but caution is not required for routine results reporting. The unhedged-assertion inventory therefore answers how often writers left out

caution where it was actually warranted, not how many assertion slots exist in the corpus.

A limited count of assertive reporting forms in the 25 zero-hedge abstracts found recurring items such as showed, revealed, observed, and highest (79 tokens in total, roughly 3.2 per abstract). These forms are reporting verbs and a superlative rather than a closed booster set. The same count was not run on the 35 hedged abstracts, so no comparison group exists and no claim is made that zero-hedge abstracts are more assertive overall. The observation is descriptive of the zero-hedge subset only: some of those abstracts use assertive wording, but binary or measured findings can also legitimately appear without hedges, and the absence of a coded hedge is not treated here as proof of absolute commitment.

## PART II: HEDGE FREQUENCY ACROSS DISCIPLINES

Cross-disciplinary comparison shows differences in hedge density, driven mainly by Veterinary Medicine's lower rate. Table 5 presents the hedge counts, false positives, unhedged assertions, and normalized frequency (hedges per 100 words) for each discipline.

**Table 5**

Hedge Frequency by Discipline

| Discipline | Abstracts | Words | Hedges | FPs | Unhedged assertions | Zero-Hedge | Hedges/ 100w |
|---|---|---|---|---|---|---|---|
| Agriculture | 20 | 4,300 | 27 | 14 | 6 | 8 | 0.63 |
| Biology | 20 | 4,509 | 28 | 32 | 4 | 5 | 0.62 |
| Veterinary Medicine | 20 | 5,060 | 10 | 12 | 4 | 12 | 0.20 |
| Total | 60 | 13,869 | 65 | 58 | 14 | 25 | 0.47 |

Table 5 shows that Agriculture had the highest hedge density at 0.63 per 100 words (27 hedges across 4,300 words), followed by Biology at 0.62 (28 hedges across 4,509 words), and Veterinary Medicine at 0.20 (10 hedges across 5,060 words). The overall corpus density was 0.47 hedges per 100 words. Agriculture's lead over Biology is marginal (0.01 hedges per 100 words), but Veterinary Medicine's density is notably lower than both, at approximately one-third of Agriculture's rate and one-third of Biology's.

The zero-hedge abstract count further sharpens this contrast. Of the 60 abstracts, 25 (41.7%) contained zero hedge tokens. Veterinary Medicine accounted for 12 of these zero-hedge abstracts (60% of VM abstracts), Biology for 5 (25%), and Agriculture for 8 (40%). The overrepresentation of Veterinary Medicine in the zero-hedge category is the primary driver of its low density. Sixty percent of VM abstracts contained no hedging devices at all, compared to 40% in Agriculture and 25% in Biology. A chi-square test of independence on the zero-hedge distribution across disciplines (discipline × zero/nonzero) was not significant ($\chi^2 = 5.07$, df = 2, $p = 0.079$), indicating that the observed disciplinary differences in zero-hedge rates do not reach statistical significance with the current sample size.

Per-abstract hedge distributions reveal substantial skew. Agriculture abstracts averaged 1.35 hedges (median = 1, SD = 1.84, range = 0–8), Biology averaged 1.40 (median = 1, SD = 1.31, range = 0–5), and Veterinary Medicine averaged 0.50 (median = 0, SD = 0.76, range = 0–3). The high standard deviation in Agriculture reflects the influence of Acs2, which contained 8 hedges alone. When Acs2 is excluded as a sensitivity check, Agriculture's density drops to 0.46 per 100 words, below Biology's 0.62, suggesting that the

Agriculture > Biology ranking is partly driven by a single outlier abstract. An exploratory Kruskal-Wallis test on per-abstract hedge rates (per 100 words) yielded $H = 7.44$, $p = .024$. Post hoc Mann-Whitney tests on the same rates give Veterinary Medicine versus Biology $p = .011$, Veterinary Medicine versus Agriculture $p = .034$, and Agriculture versus Biology $p = .72$. With a Bonferroni correction for three pairwise tests ($\alpha = .017$), only the Biology–Veterinary Medicine contrast remains significant. Agriculture and Biology are statistically indistinguishable on this measure. After correction, the only supported pairwise difference is between Biology and Veterinary Medicine; Veterinary Medicine does not differ from Agriculture at the corrected threshold. Abstracts within a program may share advisers, so observations are not fully independent, and within-discipline spread remains larger than the gaps between means. With a small, zero-inflated sample, the Kruskal-Wallis result and the pairwise follow-ups are treated as exploratory. A sensitivity check that removes all 19 instances of "potential" leaves 46 hedges. The Kruskal-Wallis contrast remains in the same direction, and Veterinary Medicine's zero-hedge share rises further. The same directional pattern holds without those tokens.

These frequency patterns partially align with, and partially diverge from, prior work. Afshar et al. (2014) found agriculture using the fewest hedges among expert research articles. Here Agriculture has the highest undergraduate-abstract density. The discrepancy likely reflects the gap between expert RAs and developing-researcher abstracts, and the specific epistemic demands of each PSAU program.

Veterinary Medicine's low density is more difficult to interpret. Many VM abstracts in this corpus reported clinical findings from animal studies (e.g.,

drug efficacy trials, disease detection, hematological profiling). Outcome-type coding across the full corpus in Part IV does not support explaining that low density by binary outcomes alone: Biology abstracts with binary results still hedge more than VM binary abstracts. The low rate is better treated as a VM pattern in this institutional corpus, consistent with departmental or supervisory practice, outcome mix, or both, rather than as a general consequence of binary evidence under Salager-Meyer's (1994) claim-responsibility framework.

The false positive distribution also varies by discipline. Biology produced 32 false positives, far more than Agriculture (14) or Veterinary Medicine (12). This is partly because Biology abstracts contain more instances of "potential" in aim statements and "indicated by" evidential pointers, constructions that look like hedges but do not express uncertainty about a claim. The higher false positive count in Biology does not necessarily mean Biology writers hedge more. It means they use more hedge-like vocabulary in non-hedging functions.

## PART III: RHETORICAL FUNCTIONS OF HEDGING DEVICES

Table 6 summarizes Step 2 outcomes for the coding pool. Of the retained hedges, all 65 were coded Epistemic Caution (EC). Politeness/Deference is null (P = 0). False positives totaled 58 (FP-Gen 33, FP-Rec 15, FP-Cap 8, FP-Proc 2). FP is not a function label; it records candidates that failed both the Working Test and the Reader Test. The direct comparison is therefore EC = 65, P = 0, FP = 58.

**Table 6**

Step 2 Outcomes: EC, P, and FP (Candidate Pool)

| Step 2 outcome | Count | Note |
|---|---|---|
| Epistemic Caution (EC) — retained | 65 | Passed Working Test |
| Politeness/Deference (P) — retained | 0 | Null in this sample |
| False positive (FP) — not retained | 58 | Failed both tests; not a function |
| Candidate decisions (EC+P+FP) | 123 | 65 retained + 58 FP |

No retained hedge was independently labeled Politeness. Under the Step 2 rule (Working Test + Reader Test), only tokens that pass at least one check enter the retained hedge set. Candidates that fail both are logged as false positives and do not count in the 65. P-only mitigation could in principle have been retained if the Reader Test held without a truth-commitment change. In this corpus no such P forms appeared (P = 0). Framing and sequencing candidates that failed both tests are included in the FP inventory (n = 58).

Most hedges in this corpus fall in implication and conclusion sentences rather than in direct results reporting. Forms such as potential and suggest/suggests/suggesting typically frame applications or interpretations. Results sentences in the same abstracts often use factive verbs (showed, revealed, demonstrated) without qualification. A formal move-structured count was not performed. This remains a qualitative pattern within the present sample.

The relationship between hedge type and function further illuminates the patterns. Modal verbs (21 instances, all EC) were used almost exclusively to express possibility about results or implications, as in “the biosynthesis.. may affect the overall development.” Lexical hedges (41 instances, all EC) were used

to qualify the strength of findings, as in “results suggest that fermented banana peel.. can significantly enhance growth.” All retained hedges were coded Epistemic Caution at Step 3. In this sample no P candidates entered the retained set. Findings framing and sequencing without mitigation were logged as false positives at Step 2 (failed both removal checks), outside the retained set. The data support a claim about the functional profile of the retained set, not a claim that writers never mitigate interpersonally or textually outside that set.

## PART IV: CROSS-DISCIPLINARY CONTRASTS

Parts I through III established frequency, function, and density differences in this corpus. Part IV asks what those differences amount to once outcome type is in view. The aim is not to treat them as stable field traits, but as patterns in these sixty ratified abstracts. Inferential tests remain exploratory (Scope and Limitations). The discussion draws on Salager-Meyer’s (1994) claim-responsibility framework and Hyland’s (2005) stance model.

**Table 7**

Outcome Type by Discipline

| Outcome Type | Agriculture | Veterinary Medicine | Biology | Total |
|---|---|---|---|---|
| Binary | 0 | 8 | 7 | 15 |
| Continuous | 4 | 4 | 2 | 10 |
| Mixed | 2 | 3 | 4 | 9 |
| Null/NS | 14 | 2 | 5 | 21 |
| Other | 0 | 3 | 2 | 5 |
| Total | 20 | 20 | 20 | 60 |

Outcome type classifies how each abstract reports its main results. Coding used the abstract’s own results language, not the title or methods alone. The five categories are as follows. Binary (15 abstracts) means presence or

absence findings or clear yes or no outcomes, such as zone of inhibition versus none, pathogen detected versus not detected, or positive versus negative diagnostic results. Continuous (10 abstracts) means graded quantities such as means, yields, rates, concentrations, or growth measures. Mixed (9 abstracts) means the abstract combines more than one outcome style, for example binary plus continuous, or continuous plus a null test. Null/NS (21 abstracts) means the abstract explicitly reports non-significance, no significant difference, or a null statistical result. Other (5 abstracts) means the results language does not fit the four boxes cleanly, for example descriptive profiles, rankings, or designs that are not clearly binary, continuous, mixed, or null. Table 7 shows the distribution across disciplines. These categories are not hedge types. They describe study outcomes so that density patterns can be read against result style. On that coding, the binary-outcome account of VM density is not supported. VM Binary abstracts average 0.38 hedges (n=8). Biology Binary abstracts average 1.57 (n=7). Same outcome class, substantially more hedging in Biology. The low VM rate is therefore better read as a pattern within this VM subset than as an effect of binary evidence as such. Agriculture accounts for 14 of 21 Null/NS abstracts, against 2 of 20 in Veterinary Medicine, and those Agriculture Null/NS texts average about 1.5 hedges. That elevation, however, is carried by Acs2 (8 hedges). Without Acs2 the remaining 13 Null/NS Agriculture abstracts average 1.0 hedge, the same as the non-Null Agriculture abstracts. Acs2 itself still matters as a single case: the writer reports that statistical significance was not observed, then hedges that the treatment could still offer potential benefits. That rescue framing matches the introduction's concern with overclaiming, but one abstract cannot ground a claim about Null/NS

writing as a class. Agriculture also has zero Binary outcomes in this sample, so discipline and study design are entangled. A formal comparison of hedge rate by outcome type across all 60 abstracts is not reported. With small cells, both the VM reading and the outcome-type patterns remain suggestive.

An informal reading of the tokens suggests hedges cluster in implication and conclusion sentences more than in direct results reporting. That pattern is strongest in Agriculture and weakest in Veterinary Medicine. A formal move analysis would be needed to quantify it. Under Salager-Meyer's (1994) claim-responsibility account, writers who hedge implications take partial distance from conclusions that go beyond the measured finding. writers who state implications flatly take fuller responsibility for those claims. In this corpus, the weaker implication/conclusion hedging in Veterinary Medicine is therefore readable as more direct claim ownership in that subset. That reading is an interpretation of the frequency pattern under the claim-responsibility lens, not a tested finding about field-wide norms. Hyland's (2005) stance model supports the same reading: fewer hedges in those moves mean less explicit softening of authorial commitment where the abstract turns from results to what the results mean.

The outcome-type data push the same point further without overclaiming causes. Binary abstracts in Veterinary Medicine average fewer hedges than Binary abstracts in Biology, so lower VM density is not explained by binary result style alone. What remains is a local pattern in this VM subset: when results are reported as presence/absence findings, these abstracts still use fewer epistemic softeners than Biology abstracts of the same outcome class. That is consistent with differential management of claim responsibility across

programs in this sample, not with a universal rule that binary evidence needs no hedge. Acs2 illustrates the opposite pole in a single Agriculture case: after reporting non-significance, the abstract still hedges residual benefits. That sequence matches the introduction's concern with overclaiming, but it remains one ratified text, not a Null/NS class effect.

The study cannot separate genuine field norms from adviser-enforced conventions. The findings describe the ratified abstracts, not individual cognition. Read as a local baseline, the corpus supports a limited claim: among these three PSAU programs, the only pairwise density contrast that survives correction is Biology versus Veterinary Medicine, and that contrast is not reducible to outcome type in the Binary cell comparison above. Whether that pattern reflects program practice, adviser feedback, or other institutional factors cannot be decided from abstract text alone; the present study maps the textual result, not its causes.

# CONCLUSIONS AND RECOMMENDATIONS

This study mapped hedging in 60 undergraduate thesis abstracts across Agriculture, Veterinary Medicine, and Biology at PSAU.

## CONCLUSIONS

This study set out to map how developing researchers at PSAU encode caution in the thesis abstract, a short genre where every claim is visible and already ratified by advisers. The picture that emerges is not a rich, field-by-field rhetoric of uncertainty. It is a constrained practice. Writers often state results with little or no softening. When they do hedge, they reach for a small lexical set and use it mainly to qualify what the findings might mean, not to elaborate stance at length. Under the coding rules used here, that work is epistemic. The abstract is treated as a place to report and to edge implications, not as a place to negotiate interpersonal face or to manage discourse for its own sake.

Disciplinary difference exists, but it is thinner than a strong "three cultures" story would predict. Veterinary Medicine is the clearest low-hedge corner of the sample. Agriculture and Biology are less distinct from each other once a single high-hedge Agriculture abstract is taken seriously as an outlier rather than as proof of a Null/NS style. The deeper point is institutional as much as linguistic. These texts are supervised products. What looks like a writer's caution profile may also be a program's tolerance for soft claims in a high-stakes summary. The data cannot name the cause. They can show the textual outcome.

The contribution is therefore a baseline, not a verdict on competence or on whole disciplines. For BAELS and related instruction, the finding matters because it shows what "hedging in science abstracts" looks like on the ground at this campus: uneven, lexically narrow, and easy to miss if teaching assumes a full academic repertoire. For research, the same baseline makes later comparison possible, against expert abstracts, other programs, or fuller thesis chapters where other functions of mitigation may have more room.

## RECOMMENDATIONS

The baseline in the conclusions is useful only if it changes what people do next. A map of uneven, lexically narrow caution in ratified abstracts does not, by itself, improve writing. It becomes useful when advisers and researchers treat it as a prompt to act on claim strength rather than on cautious wording for its own sake.

For teaching, that means dropping the idea of caution as a general polish. Hedging is a judgment about how far a sentence may go beyond the data. Advisers already see the final abstract and often shape it. The opening is to make that judgment visible in feedback, especially where the text turns from results to meaning. Students need help seeing when a finding can stand plain and when an implication needs restraint. They do not need a quota of soft wording, and they do not need to confuse move-organizing language with epistemic caution. Instruction that assumes a full academic hedging repertoire will miss what this campus sample actually shows: many abstracts soften little, and when they do, the work is concentrated and easy to misread as mere style.

For research, the same baseline is a reference point, not a finding to rest on. The open questions are causal and comparative. How much of the pattern is

program practice, how much is language resource, and how much is the mix of study designs that programs tend to produce? Abstract text alone cannot answer those questions. Designs that add other institutions, adviser accounts, draft histories from first version to final abstract, or a proficiency measure would test mechanisms this study could only name. Longer genres and expert abstracts matter for a different reason. They show whether the constrained profile seen here is mainly a product of the short, supervised summary or a wider feature of developing research writing in these fields. Method tools from this thesis can travel with that work. The priority is better questions about cause and scope, not a thicker description of the same sixty texts.

# REFERENCES


Afshar, H. S., Moradi, M., & Hamzavi, R. (2014). Frequency and type of hedging devices used in the research articles of humanities, basic sciences and agriculture. *Procedia – Social and Behavioral Sciences, 136*, 70–74. https://doi.org/10.1016/j.sbspro.2014.05.290

Biber, D., Conrad, S., & Reppen, R. (1998). *Corpus linguistics: Investigating language structure and use*. Cambridge University Press.

Crompton, P. (1997). Hedging in academic writing: Some theoretical problems. *English for Specific Purposes, 16*(4), 271–287. https://doi.org/10.1016/S0889-4906(97)00007-0

Demir, C. (2018). Hedging and academic writing: An analysis of lexical hedges. *Journal of Language and Linguistic Studies, 14*(4), 74–92. https://files.eric.ed.gov/fulltext/EJ1201945.pdf

Escalona, S. L. (2025). Hedges and boosters in college essays: A study of linguistic modulation in student writing. *International Journal of English Literature and Social Sciences, 10*(1), 7–17. https://doi.org/10.22161/ijels.101.2

Ghia, E., Petrocelli, E., & Pizziconi, S. (2022). Data possibly suggest… Hedging in second language writing: A study on advanced Italian EFL learners. *Belgrade English Language and Literature Studies, 14*, 47–72. https://doi.org/10.18485/bells.2022.14.2

Hamidi, H., Najafi Sarem, S., & Lotfi, A. H. (2025). Exploring hedging devices in scientific research papers: A content analysis study of the Medical Journal of the Islamic Republic of Iran. *Medical Journal of the Islamic*

*Republic of Iran, 39*, Article 125. https://pmc.ncbi.nlm.nih.gov/articles/PMC12584097/

Hu, G., & Cao, F. (2011). Hedging and boosting in abstracts of applied linguistics articles: A comparative study of English- and Chinese-medium journals. *Journal of Pragmatics, 43*(11), 2795–2809. https://doi.org/10.1016/j.pragma.2011.04.007

Hyland, K. (1996). Writing without conviction? Hedging in science research articles. *Applied Linguistics, 17*(4), 433–454. https://doi.org/10.1093/applin/17.4.433

Hyland, K. (1998). *Hedging in scientific research articles*. John Benjamins Publishing.

Hyland, K. (2005). Stance and engagement: A model of interaction in academic discourse. *Discourse Studies, 7*(2), 173–192. https://doi.org/10.1177/1461445605050365

Hyland, K., & Milton, J. (1997). Qualification and certainty in L1 and L2 students' writing. *Journal of Second Language Writing, 6*(2), 183–205. https://doi.org/10.1016/S1060-3743(97)90033-3

Imataki, N., & Kobayashi, Y. (2024). Analyzing disciplinary variation in abstracts: Linguistic features across scientific and technical fields. *Linguistics and Literature Studies, 12*(3), 19–28. https://doi.org/10.13189/lls.2024.120301

Khunasathitchai, K., Sittironnarit, S., Sinturat, T., & Kumdee, S. (2025). Move analysis and hedge use in the research paper abstracts of Thai EFL English-majored undergraduates. *rEFLections, 32*(2), 951–973. https://doi.org/10.61508/refl.v32i2.282512

Landis, J. R., & Koch, G. G. (1977). The measurement of observer agreement for categorical data. *Biometrics, 33*(1), 159–174. https://doi.org/10.2307/2529310

Lopez, R. M. C., Pedida, J. A., Alzate, L. J. A., & Eustaquio, M. T. L. (2025). Language of caution and certainty: Writing proficiency in hedges and boosters in student essays. *International Journal of Research and Innovation in Social Science, 9*(11), 6176–6196. https://doi.org/10.47772/IJRISS.2025.91100485

Mazidah, E. N., Cahyani, W. I., & Masruroh, L. (2022). Hedges and boosters in undergraduate thesis abstracts of Qomaruddin University students. *Journal of Education, Humaniora and Social Sciences (JEHSS), 4*(3), 1657–1663. https://doi.org/10.34007/jehss.v4i3.925

Mohammed, S. J. (2011). *Lexical hedges: A study of cultural and disciplinary variation in academic research articles* [Master's thesis, Universiti Teknologi MARA]. CORE. https://core.ac.uk/download/78277430.pdf

Ningrum, S., Puspita, H., & Mulyadi, A. I. (2024). Hedges and boosters in academic writing of ASEAN EFL learners. *Journal of English Education and Teaching, 8*(1), 202–218. https://doi.org/10.33369/jeet.8.1.202-218

Petchkij, W. (2019). Explicit teaching of hedges: Bringing hedging in academic writing into the Thai EFL classroom. *Electronic Journal of Foreign Language Teaching, 16*(1), 95–113. https://e-flt.nus.edu.sg/wp-content/uploads/2020/09/petchkij.pdf

Salager-Meyer, F. (1994). Hedges and textual communicative function in medical English written discourse. *English for Specific Purposes, 13*(2), 149–170. https://doi.org/10.1016/0889-4906(94)90013-2

Souici, F., & Slougui, D. (2022). Hedges in biology research articles: What types and frequency do Algerian authors use? *Journal of Human Sciences, 33*(4), 49–59. https://revue.umc.edu.dz/h/article/view/4015

Suratno, A., & Aydawati, E. N. (2025). Hedging strategies in non-science theses: A corpus-based analysis of discussion sections. *ETERNAL (English Teaching Journal), 16*(1), 37–52. https://journal2.upgris.ac.id/index.php/eternal/article/view/1059

Villanueva, V. S. (2014). Hedging devices in Philippine newspaper editorials. *Asian Journal of English Language Studies, 2*, 103–117. https://doi.org/10.59960/2.a5

Wang, X. (2022). *Hedging in academic writing: Cross-disciplinary comparisons in the Michigan Corpus of Upper-Level Student Papers (MICUSP)* [Preprint]. OSF Preprints. https://osf.io/preprints/edarxiv/5xj27

Xie, S., & Mi, C. (2023). Promotion and caution in research article abstracts: The use of positive, negative and hedge words across disciplines and rankings. *Learned Publishing, 36*(2), 249–265. https://doi.org/10.1002/leap.1515

# APPENDICES

## APPENDIX A: HEDGING CODEBOOK

• Lookup: Quick-Reference + Legend (p. 1). Full Codebook from p. 2. Coding sheet columns at end.

### QUICK-REFERENCE

• Context decides. A listed word is a hedge only when it works as one in the sentence.

**Types**

- Modal: may, might, could, would, should, can (possibility only, not capability or obligation).
- Lexical: suggest, indicate, appear, seem, possible, likely, potential (when they qualify a claim and pass Step 2).
- Adverbial: generally, typically, possibly, probably (scope or certainty, not procedural routine).
- Clause-level: it seems that, it is possible that, to some extent.

**Functions: EC = uncertainty about a claim's truth. P = interpersonal softening/deference. FP = fails Working Test and Reader Test (not a function).**

**Core rules:** suggest/indicate finding→hedge, recommendation→FP-Rec; can/may possibility→hedge, capability→FP-Cap; generally/typically claim-limit→hedge, procedural→FP-Proc; appear/seem result-uncertainty→hedge, location→FP-Gen.

**Steps 1–3: Step 1 wide-net candidates. Step 2 = Working Test + Reader Test (remove the device): truth-commitment changes → EC; commitment same but claim more imposing/unattributed/confrontational toward readers or prior work → P; neither → FP. Step 3 = grammatical type on retained P and EC only. Functions = P and EC only. FP is a reject record, not a function.**

### QUICK-REFERENCE LEGEND

• Highlights = type background on the token. Colored text labels = function. Grey = unhedged verb. Strike + grey code = FP.

| TYPE (Highlight) | FUNCTION (Text) |
|---|---|
| Modal Verbs | EC Epistemic Caution |
| Lexical Hedges | P Politeness / Deference |
| Adverbial Hedges | — |
| Clause-Level Hedges | |
| **UNHEDGED ASSERTIONS (Grey Highlight, Optional)** R = Result, G = Generalization, C = Causal. Bracket the subtype beside the mark. | **FALSE POSITIVE (Strikethrough)** No highlight. Strike through the token and write the reason code in grey text. |

### COMBINED EXAMPLE

"The results suggested (EC) that extracts are potential (EC) alternative stains. PCR ~~can~~ (FP-Cap) also detect low concentrations."

Type highlight = grammatical type. Orange text = function (EC). Strike + grey code = false positive. Here both suggested and potential are retained Lexical EC; can is FP-Cap (capability).

- *Full rules in Full Codebook (from next page).*

**FULL CODEBOOK**

**PURPOSE**

- Coding scheme for the PSAU undergraduate thesis abstract corpus, adapted from Hyland (1998) as type instrument.

**STEP 1, STEP 2, STEP 3, AND WORKING TEST**

- Step 1 (wide net). After the holistic read, mark all possible hedge candidates. This step does not count retained hedges.

- Step 2 (function by paired removal checks). For each candidate, remove the device and ask what changes. Working Test: commitment to the truth of the proposition changes → retain as EC. Reader Test: truth-commitment stays the same, but the claim becomes more imposing, unattributed, or confrontational toward readers or prior work → retain as P. Neither changes → not a hedge; log as FP (not a function). Both may hold on one token (EC+P).

- Working Test and Reader Test are the Step 2 function gate. Capability, recommendation, pure procedure, and topic-naming typically fail both checks and are logged as FP.

- Step 3 (type). On retained P and EC tokens only, assign grammatical type (modal, lexical, adverbial, clause-level). Function is already fixed at Step 2 (Working Test and/or Reader Test). FP is not labeled at Step 3.

- Order. Coding order: full abstract → Step 1 → Step 2 (Working Test + Reader Test) → Decision Rules → Step 3 type on retained P and EC → Coder Notes. Deontic advice is never retained (FP-Rec).

**WORKING EXAMPLES: STEP 2 (WORKING TEST + READER TEST) AND STEP 3 TYPE**

- Function labels apply to retained hedges after Step 2. Step 2 = claim mitigation for findings, not advice. Working Test supports EC when certainty changes.

- EC (Working Test): "The treatment may reduce infection rates." Removing may raises truth-commitment → EC.

• FP-Cap: “PCR can detect low concentrations.” can = capability, not uncertainty about a finding.

• P (Reader Test): “As suggested by previous researchers, the treatment is effective.” Removing the attribution leaves truth-commitment intact but makes the claim more unattributed/imposing toward prior work → P.

• P + EC: Both Working Test and Reader Test hold on one token → EC+P (or dominant label plus dual note in Coder Notes).

• FP: “In conclusion, T3 produced the highest yield.” Removing the frame changes neither truth-commitment nor interpersonal imposition toward readers/prior work → FP. Not a function label.

• Frame + separate EC: “It should be noted that the results may indicate growth.” Code may via Working Test as EC. The frame alone fails both tests and is not a second hedge.

• Summary: Step 1 is wide. Step 2 assigns function with the Working Test (EC) and Reader Test (P). Neither → FP (not a function). Step 3 assigns type to retained hedges only. In this corpus all 65 retained tokens were EC. No P forms appeared. FP cases are inventoried separately.

**HEDGE TYPE (PRIMARY COLORS)**

Highlight each hedge token in the color matching its grammatical form.

| Color | Type | Definition | Highlighted Example |
|---|---|---|---|
| | Modal Verbs | Expresses possibility, probability, or a condition under which something may happen. | “This **may** indicate disease resistance.” |
| | Lexical Hedges | Verbs, adjectives, or nouns that make a claim less certain or more cautious. | “Data **suggests** a correlation.” |
| | Adverbial Hedges | Adverbs that qualify how broadly, how often, or how certainly a claim applies. | “Plants **generally** respond to drought.” |

| Color | Type | Definition | Highlighted Example |
|---|---|---|---|
| | Clause-Level Hedges | Clause-level phrases framing a claim as tentative. | "It seems that the pathogen is host-specific." |

*Exclusions (capability, recommendation, procedural norm, location) are in the Decision Rules table.*

• imply / implies / implying = suggest. • can be concluded / derived / stated = inferential hedge (not capability can). • potentially = adverbial. • potential + noun: aim/methods topic-naming = FP-Gen. Results/conclusion qualifying a finding = Lexical EC. If removing potential changes commitment to a proposition, code Lexical EC.

## HEDGE FUNCTION (COLORED TEXT LABELS)

| Label | Function | Definition | Marked Example |
|---|---|---|---|
| EC | Epistemic Caution | Expresses uncertainty about a claim's truth. | "The treatment may reduce infection rates. (EC)" |
| P | Politeness / Deference | Shows respect to readers or prior research. | "As suggested by Hyland (1998). (P)" |
| FP | (not a function) | Fails Working Test and Reader Test | "In conclusion, T3 produced the highest yield." |

## DECISION RULES FOR AMBIGUOUS CASES

Apply before defaulting to the Working Test.

| Token | Hedge When | Not a Hedge When |
|---|---|---|
| **suggest / indicate** | Expresses uncertainty about a finding or conclusion | Introduces a recommendation or action |
| **suggest / indicate + need for** | Says the evidence suggests a need may exist | Says what should be done or is needed |
| **indicated by** | Introduces a conclusion drawn from evidence | Points to evidence for a claim already stated |
| **indicate / indicates that** | Introduces an inferential conclusion drawn from analysis | Used as a synonym for showed/demonstrated (factive reporting) |
| **indicating (participial)** | Introduces an inference beyond the operational definition of the result | Restates the operational meaning of statistical significance |
| **can / may** | Expresses possibility | Expresses capability |
| **generally / typically** | Qualifies scope | Describes a procedural norm |
| **appear / seem** | Expresses result uncertainty | Indicates location |

***General exclusions:*** *ability, future procedure, and procedural necessity (a step required by the procedure itself) are never coded as hedges.*

*Quick test for* ***suggest/indicate****: "Might this be true?" = hedge. "Should this be done?" = FP-Rec.*

## MARKING FALSE POSITIVES

Strike through a hedge-like token that fails the Working Test. Write the reason code beside it in grey text.

| Code | Reason | Marked Example |
|---|---|---|
| FP-Cap | Capability: describes what something is able to do, not uncertainty about whether it will happen. Distinct from inferential "can be concluded / derived / stated / attributed to / seen as," which stays a hedge. | "PCR ~~can~~ (FP-Cap) detect low concentrations." |
| FP-Rec | Recommendation: expresses what should, needs to, or ought to be done, not uncertainty about a claim. | "We ~~suggest~~ (FP-Rec) further research." |
| FP-Proc | Procedural norm: describes a routine, standard, or established procedure rather than uncertainty about a claim. | "Samples were ~~generally~~ (FP-Proc) stored at 4°C." |
| FP-Loc | Location: indicates physical or textual location. | "The results ~~appear~~ (FP-Loc) in Table 1." |
| FP-Gen | General exclusion: doesn't fit the codes above. Use for expressions that look hedge-like but don't express uncertainty (e.g., naming a research topic, pointing to evidence). Note the specific reason in Coder Notes. | "...demonstrated notable antioxidant activity, as ~~indicated~~ (FP-Gen) by the reduction of DPPH radicals." |

## UNHEDGED ASSERTIONS (WARRANTED CAUTION, NEUTRAL GREY)

- Grey-highlight the main assertive verb; place [G] or [C] beside it (same placement as (EC) on hedges).
- Mark only claims where caution was warranted but absent.
- [G] = generalization or evaluative ranking beyond the data.
- [C] = cause–effect or impact stronger than the design supports.
- Do not mark bare trial results, measured findings, or procedural statements.
- Record the full clause in the coding sheet.

| Color | Subtype | Definition | Highlighted Example |
|---|---|---|---|
| | Result Claim [R] — Not markable (bare results). | Not markable. | "was [G] the most compelling..." (mark) vs "Yield higher ($p < .05$)" (omit). |
| | Generalization [G] | Inference or ranking beyond the data. | "This led [G] to the conclusion that all farms were free of it." |

| Color | Subtype | Definition | Highlighted Example |
|---|---|---|---|
| | Causal Claim [C] | Cause–effect stated as established. | "Extracts produce [C] comparable staining because they contain similar compounds." |

**CODING SHEET COLUMNS**

| Abstract ID | Sentence # | Clause | Token | Reason Code | Coder Notes |
|---|---|---|---|---|---|

| Abstract ID | Sentence # | Clause | Subtype | Reverse Test (Y/N) | Coder Notes |
|---|---|---|---|---|---|

| Abstract ID | Sentence # | Clause | Hedge Token | Type | Function | Word Count | Coder Notes |
|---|---|---|---|---|---|---|---|

• Highlighted abstract = primary record. Spreadsheet = tallying and analysis only.

• *Codes: Acs1–Acs10, Aas1–Aas10 (Agriculture). VM1–VM20. B1–B20 (Biology).*

• *Unhedged rows separate from hedge tallies. [G]/[C] only for warranted-but-absent caution.*

• *Optional Rejected Candidates tab for FP traceability.*

# APPENDIX B: PILOT SAMPLE ANALYSIS AND CODING VERIFICATION

Four abstracts drawn from the pre-2023 pool were coded before the main corpus. The goal was to test the scheme, settle borderline cases, and see whether zero-hedge abstracts were common enough to shape the study design. Each abstract is given below with its coding summary.

*Sample 1: Biology*

A field and laboratory monitoring was conducted at the Bureau of Fisheries and Aquatic Resources Regional Fish Health Laboratory 3 (BFAR RFHL3) and Zambales, Philippines to compare the effectiveness of Nested Polymerase Chain Reaction and Micro-Insulated Isothermal PCR (Pocket) in monitoring Acute Hepatopancreatic Necrosis Disease.

A total of 33 samples were used in this study to compare the results of Nested Polymerase Chain Reaction and Micro-Insulated Isothermal Polymerase Chain Reaction (POCKIT) in monitoring Acute Hepatopancreatic Necrosis Disease-causing Vibrio parahaemolyticus. The study used collected shrimp samples, pre-detected samples, and positive and negative standards for the detection of Acute Hepatopancreatic Necrosis Disease-causing Vibrio parahaemolyticus on shrimp samples; bacterial culture for vibrio sp. was also conducted as well as dna sequencing for the confirmation of the results. The result showed negative results on all the shrimp samples that led to the conclusion that all shrimp farms were free from Acute Hepatopancreatic Necrosis Disease-causing Vibrio parahaemolyticus.

**Hedge Count: 0**

Analysis: The abstract relies entirely on factive verbs (conducted, showed, led to the conclusion). The final claim is presented as categorical with no qualification.

**Key Excerpt:** "The result showed negative results that led to the conclusion that all shrimp farms were free..."

*Sample 2: Veterinary Medicine*

The study was conducted to determine and compare the staining effect of C. longa, I. batatas and M. acuminata aqueous extracts for gonad examination of Nile tilapia. Twelve Nile tilapia fingerlings, with approximately two and one-half to four-inch size, were divided into four treatments with three replicates each. Treatment 1 was assigned as the positive control, which were stained with malachite green, while Treatments 2, 3 and 4 were stained with C. longa, I. batatas and M. acuminata extracts, respectively. The prepared slides of stained gonads were randomly distributed on the microscopes using completely randomized design (CRD). Blind examination and evaluation by five faculty members was done on the stained unlabeled tissues under the microscopes. The tissues were evaluated according to five parameters: hue, saturation, contrast, clarity and overall appearance. A scoring scheme was followed, i.e. 1 to 2 – very poor; 3 to 4 – poor; 5 to 6 – average; 7 to 8 – good; and 9 to 10 – excellent. Data gathered were subjected to one-way analysis of variance (ANOVA). The results suggested that C. longa, I. batatas and M. acuminata extracts are potential alternative stains for gonadal examination of Nile tilapia with M. acuminata as the most preferred.

**Hedge Count: 2**

Analysis: This abstract demonstrates awareness of claim management. Both hedges serve to qualify the final conclusion.

**Coded Instances:**

"The results suggested that..." (Lexical Verb - Epistemic Caution)

"...extracts are potential alternative stains..." (Lexical Adjective - Epistemic Caution)

**Excluded:** "approximately two and one-half" (Procedural measurement, not epistemic).

*Sample 3: Agriculture (Animal Science)*

The study was conducted for 7 days from July 07 to July 13, 2022 at [REDACTED]. The general objective of the study was to determine the acceptability and palatability of duck ham subjected to different curing process. Specifically, it aimed to find the sensory evaluation: appearance; aroma; taste/ flavor; texture; and over-all acceptability. A total of eight (8) Muscovy ducks were used in this study. The study used Hedonic Likert Scale and was subjected to Desriptive analysis. Factor A = Type of curing and Factor B = Spice rub. The treatments used in this study are the following. $A_1$ $B_1$ - Dry curing + Oregano, $A_1$ $B_2$ - Dry curing + Lemon grass, $A_2$ $B_1$ - Liquid curing + Oregano and $A_1$ $B_2$ - Liquid curing + Lemon grass. The sensory evaluation was conducted at Pampanga State Agricultural University, PAC, Magalang, Pampanga. Ten (10) college students, ten (10) teachers and ten (10) people who don't eat duck were selected as respondents to evaluate the sample treatments. The data collected from the sensory evaluation were analysed using descriptive statistics.

Results of the study showed that Comparison on Curing Methods Using Oregano - dry cured duck ham had the highest rating with regards to appearance, texture and overall acceptability. While liquid cure duck ham got the highest rating in terms of aroma and taste/flavor. Comparison on Curing Methods Using Lemon Grass - liquid cured duck ham had the highest rating in

terms to appearance, aroma, taste/flavor, texture and overall acceptability. Comparison of Spices on Dry Curing (Smoking) - dry cure duck ham smoked in lemon grass had the highest rating in terms of appearance, aroma, taste/flavor, texture and overall acceptability. Comparison of Spices on Liquid Smoke - liquid cured duck ham smoked in lemon grass had the highest rating in relation to appearance, aroma, taste/flavor, texture and overall acceptability. Descriptive rating revealed that all the treatment with regards to appearance, aroma, taste/flavor, texture and overall acceptability were rated moderately like.

**Hedge Count: 0**

Analysis: The abstract uses factive reporting verbs (showed, revealed). Sensory evaluation ratings are reported as absolute data points. The term "moderately like" describes a Likert scale variable, not an epistemic qualification.

**Key Excerpt:** "Results of the study showed that Comparison on Curing Methods..."

*Sample 4: Agriculture (Crop Science)*

The study was conducted from June to July 2022 to determine the influence of the different sources of nutrients to rice straw mushrooms. The field experiment was conducted in [REDACTED]. The experiment was laid out using Complete Randomize Design (CRD), with four treatments replicated four times. The treatments were as follows: $T_0$- Untreated banana leaves substrate submerged in 55 gallon of water (Control); $T_1$- Banana leaves substrate submerged in mixture of 100 grams of Urea and 55 gallons of water; $T_2$- Banana leaves substrate submerged in mixture of 250 ml of fish amino acid and 55

gallons of water; and, $T_3$- Banana leaves substrate submerged in mixture of 250 ml of fermented plant juice and 55 gallons of water.

Results of the study revealed significant differences in the following parameters: average weight of button harvested, number of buttons harvested, average yield per harvest, and biological efficiency of rice straw mushroom. However, insignificant differences were observed on the days of pinhead appearance and days of harvesting. The highest ROI was obtained in $T_1$- Banana leaves substrate submerged in a mixture of 100 grams of Urea and 55 gallons of water with every peso of investment 1.01 pesos of net income return.

**Hedge Count: 0**

Analysis: Statistical significance is reported as fact (revealed significant differences). No devices are used to qualify the biological efficiency or yield claims.

**Key Excerpt:** "Results of the study revealed significant differences... The highest ROI was obtained..."

## General observation from the pilot

Three of the four pilot abstracts contained no coded hedges. The pattern pointed to a high base rate of zero-hedge writing in the local undergraduate abstract genre and confirmed that a descriptive baseline study was warranted. The same decision rules and Working Test were carried forward into the main coding of the 60-abstract corpus.

# APPENDIX C: CODED CORPUS WITH HIGHLIGHTS

This appendix presents the 60 abstracts with all coding marks applied. Type marks = color highlight on hedge tokens (blue = Modal Verbs, red = Lexical Hedges, yellow = Adverbial Hedges, green = Clause-Level Hedges). Function marks = function label in colored text (EC = orange, P = purple). Discourse connectors (Hence/Thus) are tagged even when not hedges. Function labels in this appendix are only EC and P. False positives = strikethrough + reason code in grey (e.g. FP-Rec = recommendation, not a function). Unhedged assertions = grey highlight + [G]/[C] bracket (warranted-but-absent caution only).

## Biology (B1–B20)

### B1 · 258 words · 2 hedges · 0.78 per 100 words

Plant Growth Promoting Bacteria (PGPR) holds significant agricultural importance as a widely used microbial biofertilizer in various agricultural practices. Among the commonly prevalent PGPR are the Bacillus spp., specifically Bacillus subtilis which have been found to act as bio-fertilizers, biostimulants, and biological control agents. In this study, the growth promoting ~~potential~~ (FP-Gen) of Bacillus subtilis BIOTECH 1679 was assessed under greenhouse conditions. Results ~~indicated~~ (FP-Gen) that Bacillus subtilis formulation (9.0 x $10^8$ CFU/mL) of 10 g/L (T1), 30 g/L (T2), and 50 g/L (T3) concentrations exhibited a notable increase in plant growth parameters. Based on statistical analysis using the Kruskal-Wallis test, Welch One-way Analysis of Variance, and One-way ANOVA, the three varying concentrations of B. subtilis formulations have no significant differences in their effects on the

plant growth parameters including plant height, number of leaves, length of leaves, and width of leaves of onion plants (p-value ≥ 0.05). They also presented no significant difference in their effects with the positive control, Nitrogen-Phosphorus-Potassium (NPK 20:20:20) (p-value ≥ 0.05). This implies (EC) that the use of low concentration of B. subtilis BIOTECH 1679 are as effective in promoting the growth of onion. Welch ANOVA revealed that different treatments showed significantly larger leaf width (p-value ≤ 0.05). However, One-way ANOVA revealed that B. subtilis BIOTECH 1679, regardless of the concentration used, demonstrated shorter root length as compared with that of NPK (20:20:20) (p-value ≤ 0.05). The results provided valuable insights into the potential (EC) of B. subtilis as a sustainable solution for managing onion supply demands, contributing to eco-friendly and alternative strategies in agriculture from local to global use.

**B2 · 189 words · 0 hedges · 0.0 per 100 words**

This study was conducted at the Research Extension and Technology (RET) and Microbiology Laboratory of the College of Arts and Sciences of Pampanga State Agricultural University from November 2018 - January 2019. The study aims to determine the antibacterial activity of the Partial Fractions of Balitbitan (Cynometra ramiflora) against selected bacterial pathogens, Staphylococcus aureus and Escherichia coli. In the preliminary Kupchan method of Partitioning, the ethanolic extract of Balitbitan (Cynometra ramiflora) was partitioned to three solvents which are Hexane, Methanol and DCM and concentrated using a rotary evaporator at low temperature (30°C) and pressure. In the antibacterial testing, the paper disc diffusion method is used in order to ~~indicate~~ (FP-Proc) the zone of inhibition in the evaluation of Antibacterial

~~potential~~ (FP-Gen) of Balitbitan on Mueller Hinton Agar. The treatment evaluated as follows: T1 (Balitbitan extract in Methanol), T2 (Balitbitan extract in Hexane) and T3 (Balitbitan extract in DCM). The result of antibacterial testing revealed that the Partial Fraction of Balitbitan (Cynometra ramiflora) did not possess the property to inhibit the growth of Staphylococcus aureus and Escherichia coli. There was no zone of inhibition when treated with the partial fraction of Balitbitan.

**B3 · 142 words · 1 hedges · 0.7 per 100 words**

Clean water resources are essential for sustaining life, yet their scarcity and contamination pose significant global challenges, particularly in remote communities. This study addresses the pressing issue of water potability in [REDACTED], focusing on the detection of contaminants and pollutants that threaten human health. Water and soil samples were collected from the community, and heavy metal analysis was conducted to detect lead (Pb) and mercury (Hg). Microbial testing using Most Probable Number (MPN) was performed to assess the presence of bacterial coliform and Escherichia coli (E. coli) in water samples. Results revealed the presence of bacterial coliform and E. coli in both the main water source and communal water, indicating (EC) unsafe conditions for water consumption for November and December. In addition, lead levels were within safe limits, while mercury levels in soil exceeded recommended guidelines, raising further concerns about environmental health.

**B4 · 231 words · 3 hedges · 1.3 per 100 words**

This study evaluated the acute toxicity and antitussive activity of Mimosa pudica (Makahiya) ethanolic extract in ICR mice to assess its safety and ~~potential~~ (FP-Gen) as a natural antitussive agent or cough suppressant. Following OECD 423 guidelines, mice were administered 400 mg/kg and 2000 mg/kg of the extract and monitored for 14 days for signs of toxicity, changes in body weight and mortality. No mortality, distress, or significant weight changes were observed, ~~indicating~~ (FP-Gen) that Mimosa pudica extract is non-toxic up to 2000 mg/kg. For the antitussive assessment, cough was induced using 7.5% citric acid aerosol, and the effects of extract doses (100 mg/kg, 200 mg/kg, and 400 mg/kg) were compared with a negative control (distilled water) and a positive control (Lagundi capsule, 25 mg/kg). A dose-dependent reduction in cough frequency was observed: 100 mg/kg reduced coughs by 37.86%, 200 mg/kg by 60.72%, and 400 mg/kg by 70.00%. The 400 mg/kg dose had the strongest effect among the treatment groups and was only 11.42% less effective than Lagundi, which achieved an 81.42% reduction. These results suggest (EC) that Mimosa pudica has promising antitussive properties and a wide safety margin, supporting its potential (EC) as an accessible herbal alternative for managing cough. The presence of bioactive compounds such as flavonoids and saponins may (EC) contribute to its therapeutic effects. Further studies are ~~recommended~~ (FP-Rec) to confirm its long-term safety, refine dosing, and better understand its mechanisms of action.

**B5 · 245 words · 1 hedges · 0.41 per 100 words**

Throughout the years, foodborne pathogens have posed a significant threat to the health and safety of many. With the ongoing adversity brought by

antimicrobial resistance, other antibacterials are necessary. In the current study, the antibacterial and antioxidant ~~potential~~ (FP-Gen) of Copper oxide nanoparticles (CuO NPs) were evaluated. The green amaranth plant (Amaranthus viridis) leaves were used to synthesize the green CuO NPs. Successful formation of CuO NPs was visually confirmed through color change, through Fourier Transform Infrared Spectroscopy, and through the absorbance peak at 230nm on the UV-Vis spectrophotometry. Greenly synthesized CuO NPs had an average size of 89.3 nm in the shape of ellipsoids when characterized by Scanning Electron Microscope and ImageJ imaging software. Through DPPH assay, the antioxidant activities of the CuO NPs were determined to have an $IC_{50}$ value of 853µg/mL. Moreover, using growth-kinetic assay, antibacterial activities of the different concentrations of CuO NPs were determined and found to be statistically significantly different ($p < 0.05$) from each other for the Staphylococcus aureus, Salmonella enterica serovar Typhimurium, Shigella sp., and Bacillus cereus, ~~indicating~~ (FP-Gen) a dose-dependent response. In contrast, no significant difference ($p > 0.05$) was observed for the Enteropathogenic Escherichia coli (EPEC). Minimum Inhibitory Concentration (MIC) values were determined to be 1000 µg/mL except for EPEC, with MIC value at 500 µg/mL. However, Minimum Bactericidal Concentration (MBC) at 2000 µg/mL was observed for all test organisms. The study offers novel findings in the field of green amaranth synthesized nanoparticles as potential (EC) antioxidant and antibacterial sources.

**B6 · 232 words · 0 hedges · 0.0 per 100 words**

The study ~~would~~ (FP-Gen) be on developing and optimizing eco-bricks using a mixture of SUPs and UCO developed through trial and error. SUPs recovered

from various donors from the [REDACTED] and UCO. The preliminary trials established that molding faces some difficulty with increased ratios of UCO to SUPs. Thus, in the optimization process, varying ratios of UCO was [G] used, starting with UCO to SUPs 1:1 as treatment 1, a decreasing ratio of UCO to SUPs. At equal parts by weight of UCO and SUPs, optimization started at Treatment 1 with a 1:1 ratio. Treatments 2–5 proceeded similarly, increasing the ratio of UCO to SUPs. Density measurements, water absorption, and drop tests were all performed and observed. A self-testing protocol based on current literature was used. The density was calculated relative to mass and volume; however, the water absorption test yielded inconsistent findings. A drop test was also performed, which revealed that the eco-bricks from all treatments were significantly damaged at the height of one meter, making them a weak, unsuitable material to use in building walkways. This study is essential for getting insights into eco-brick manufacture because the workability of SUPs and UCOs as raw materials is proven. The research findings do not apply to buildings. Therefore, further research ~~will~~ (FP-Gen) aid in underlining the importance of optimizing the ingredient ratios to acquire desirable quality eco-bricks for sustainable construction applications.

### B7 · 266 words · 3 hedges · 1.13 per 100 words

The study ~~generally~~ (FP-Proc) aimed to evaluate the embryo toxicity, lethality, and teratogenicity of Ziziphus talanai-infused nanozeolite pellets on Danio rerio was performed at the Biology and Chemistry Laboratory of Pampanga State Agricultural University from October 2023 to January 2024. Various treatments in the formulation of pellets and laboratory experiments were set up using Completely Randomized Design (CRD). The study showed that cassava

starch as combined with Ziziphus talanai-infused nanozeolite at 1:9 ratio improved the water-holding capacity of derived pellets, thus making a remarkable choice as a binder. The hatching and survival of Danio rerio embryos were enhanced by 200 ppm concentration of Z. talanai-infused nanozeolite pellet suspension while demonstrating lower toxicity compared to other concentrations tested. Based on the Acute Toxicity Index, the median lethal concentration ($LC_{50}$) in D. rerio embryos at 96 hpf (hours post-fertilization) showed that Z. talanai-infused nanozeolite pellet at a concentration of 49.79 µg/mL falls within the 'slightly toxic' classification, thus demonstrating that nanozeolite enhances the delivery of compounds within Z. talanai extract. Coagulation of embryos was primarily observed within 400 ppm to 1,000 ppm concentrations of Z. talanai-infused nanozeolite pellet suspensions under 40x magnification, suggesting (EC) that it affects early embryonic development and prevents the progression of other lethality markers, such as the absence of somite development, non-detachment of the tail, and absence of cardiac activity. Teratogenic effects on the body length of D. rerio hatchlings were observed, inferring (EC) that such observation was induced by the Z. talanai-infused nanozeolite pellet suspension. Hence, the biosynthesis of Z. talanai-infused nanozeolite pellets is feasible and may (EC) affect the overall development of D. rerio embryos.

**B8 · 241 words · 2 hedges · 0.83 per 100 words**

Recently, nanotechnology has emerged as a rapidly growing field with various biomedical applications. In this study, silver nanoparticles (AgNPs) were biosynthesized using the phytopathogenic fungus Colletotrichum spp. isolated from infected tamarind leaves, and its antibacterial action was assessed against

enteric pathogens found in broiler chicken (E. coli, S. typhi, Shigella spp.). The bio-reduction of AgNPs was characterized by visual inspection of color change (clear to brown), followed by the analysis of UV-visible spectroscopy which confirms the successful production of AgNPs (strong absorption peak monitored at 420nm). The SEM photomicrographs revealed that the nanoparticles were primarily composed of silver ions with sizes ranging from 66.96nm to 89.29nm. The antibacterial ~~potential~~ (FP-Gen) of fungus-mediated AgNPs was distinguished by performing the agar well diffusion method and determining minimum bactericidal concentration (MBC) and minimum inhibitory concentration (MIC) via resazurin and MTT assays. The antibacterial activity of AgNPs was found to outstandingly inhibit all bacterial strains in a concentration-dependent manner, with the largest mean zonal inhibition of 16.1mm at 6g. The MBC and MIC were found to be in the range of <0.5g to 0.5g, and 0.5g to 1g, in both assays, respectively. The statistical analysis showed that there was a significant difference between the concentrations (p-value: 0.000) and no significant difference when compared to traditional antibiotics (p-value: >0.05). Finally, the biosynthesized AgNPs successfully inhibited the common pathogens found in chicken. Hereafter, its potential (EC) to control bacterial infections caused by multi-drug resistant pathogens could (EC) also be considered.

**B9 · 212 words · 0 hedges · 0.0 per 100 words**

This study aimed to identify the histoprotective ~~potential~~ (FP-Gen) of Balakat extract (Ziziphus talanai) against monosodium glutamate effect to the specific vital organs of Mus musculus through observing the lesions / aberrations to the heart, liver and kidney. Each replication and treatments were observed under

the compound microscope in a 100x and 400x objective. All mice treated with pure leaf extract of Z. talanai, monosodium glutamate, distilled water and L-Taurine showed a significant effect (lesions / aberrations) to the cytoarchitecture of liver and kidney. However, heart did not show any significant effect (lesion / aberration) to its cytoarchitecture. Interestingly, all histopathological observation under the combined treatments of Z. talanai and L-Taurine, Z. talanai and monosodium glutamate showed a significant reduction of lesions / aberrations score to the liver and kidney compare to the pure treatments. However, the heart didn't respond or show any significant changes regardless of the treatment. Based on the results of this study, pure MSG and pure ethanolic leaf extract of Z. talanai caused [C] the lesions / aberrations to the liver and kidney. However, treatments with combination of MSG and pure ethanolic leaf extract of Z. talanai extract results for reduction of lesions / aberrations to the organ cytoarchitecture. Lesions / Aberrations that were observed under the microscope are cell degeneration, glomerular atrophy and glomerular hypertrophy.

**B10 · 302 words · 1 hedges · 0.33 per 100 words**

It is well recognized that heavy metals harm human health through the food chain. The danger to human health associated with consuming fish contaminated with heavy metals were assessed. This study aimed to determine the presence of lead in the flesh part of selected fish species, Common carp (Cyprinus carpio), Roho labeo (Labeo rohita), and Silver carp (Hypophthalmichthys molitrix), in addition to determine the Knowledge, Attitude, and Practices (KAP) of fishers along the Pampanga River. A test method used is Inductively Coupled Plasma — Optical Emission Spectrometry

(ICP-OES) in identifying heavy metals in biota. The self-administered questionnaires were distributed to 20 respondents, questionnaires compromised of socio-demographic characteristics and the KAP's of fishers along the Pampanga River. Data analyses were done using SPSS (version 23). Data were explored using descriptive statistics. The results show that lead in any of the selected fish, with a <0.013 mg/kg result is below the FAO and WHO's allowed limits of 2 mg/kg and 0.5 mg/kg, respectively. Based on the result from KAP Analysis, knowledge scored M=1.7200 with SD equivalent to 0.10052 which is interpreted as poor knowledge. The attitude score M=3.1150 and SD=0.093330 which means a moderate attitude and lastly, practices score M = 2.8850 and SD = .13485 which is equivalent to have good practices. Thus, the researcher provided a policy proposal to focus on issues that needed to be addressed in the community. In terms of practices, proper waste management ~~should~~ (FP-Rec) be established to avoid the high danger of heavy metal contamination. Certain local habits, such as waste dumping and other communal activities, may (EC) contribute to the spread of heavy metals. As a result, some community or local government-adopted measures are ~~proposed~~ (FP-Gen). These precautions are intended to prevent heavy metal contamination, particularly in the river system and the aquatic life in the area.

**B11 · 256 words · 1 hedges · 0.39 per 100 words**

Currently, Central Luzon together with Calabarzon are regions that hold a pivotal position in the Philippines' fishing industry. However, due to their adjacency to [REDACTED], which is highly polluted due to anthropogenic activities, these regions become vulnerable to bacterial infestation. Mycobacteriosis is a significant concern in cultivated milkfish, impacting both

fish health and aquaculture productivity is reported in [REDACTED] and in parts of Central Luzon. Hence, this study aimed to investigate the presence of potential (EC) mycobacterial infections in cultivated milkfish collected from fish-supplying areas of [REDACTED]. Using the Ziehl-Neelsen stain for acid-fast bacteria detection, along with biochemical testing methods such as the catalase test and assessing growth in the presence of the Salt Tolerance Test, revealed the presence of acid-fast bacteria in the tissues of infected milkfish, confirming the occurrence of mycobacteriosis infection. Additionally, biochemical testing, specifically the catalase test, was utilized to identify the catalase enzyme activity characteristic of mycobacteria, catalase test positive shows the result of the formation of bubbles. Furthermore, the growth of mycobacteria in the presence of the Salt Tolerance Test was assessed, showing the ability of these bacteria to thrive in high-salt environments, which is crucial information for understanding their survival and spread in aquaculture settings. Overall, the combination of Ziehl-Neelsen staining, biochemical testing like the catalase test, and evaluating growth in the presence of the Salt Tolerance Test provides [G] a comprehensive approach for diagnosing and studying mycobacteriosis in cultivated milkfish, offering insights that ~~can~~ (FP-Cap) guide effective management strategies to control the spread of this disease in aquaculture system.

**B12 · 149 words · 1 hedges · 0.67 per 100 words**

Antimicrobial resistance is a major threat to public health worldwide, making infections difficult to treat. This problem calls for developing novel and alternative strategies to attenuate microorganisms without inducing resistance. Hence, lactic acid bacteria (LAB) have emerged as a focal point due to

their potential (EC) quorum quenching (QQ) activities. In this study, lactic acid bacteria were obtained from a parallel study, and the antimicrobial activity of the top 10 isolates was [C] evaluated through spectrophotometric and turbidimetric analyses for the initial screening of gut isolates. Nine out of ten isolates exhibit quorum quenching activities by inhibiting the swarming motility of Pseudomonas aeruginosa. The metabolic compounds of the isolates were subjected to initial characterization using enzyme and pH neutralization, which revealed that organic acids were responsible for the inhibitory activity. The inhibition activity of LAB isolates against the quorum sensing-mediated swarming motility in P. aeruginosa was due to strong antibacterial activity.

**B13 · 148 words · 0 hedges · 0.0 per 100 words**

Adenium obesum, also called Bangkok Kalachuchi, is a succulent shrub, and a member of the Apocynaceae family, one of the most abundant sources of phytochemicals, including alkaloids, steroids, saponins, glycosides, anthraquinones, tannins, and flavonoids, and has a lot of promise for use in pharmaceutical and horticultural products. In this study, the researcher investigated the antibacterial property of Bangkok Kalachuchi (Adenium obesum) ethanolic leaf extract against standard tester strains of Staphylococcus aureus (ATCC 25923) and Escherichia coli (ATCC 25922) using disc diffusion method. The result of this study showed that Adenium obesum ethanolic leaf extract has antibacterial property against Staphylococcus aureus (ATCC 25923) at 1,000 µg/disc and 10,000 µg/disc concentrations and none against Escherichia coli (25922). Results also showed that the antibacterial property of Bangkok Kalachuchi (Adenium obesum) against Staphylococcus

aureus (ATCC 25923) are significantly different at various concentration ~~indicating~~ (FP-Gen) that its antibacterial property is dose dependent.

**B14 · 242 words · 3 hedges · 1.24 per 100 words**

The emergence of infectious diseases driven by pathogenic microbes prompted the need for the discovery of more effective and affordable medicines. In finding alternative agents that helps combat microbial diseases, studies revealed that several plants possess pharmaceutical activities. Paragis (Eleusine indica L.) is a difficult weed in turf grass systems of the Poaceae family cultivated throughout the world. The information collected reveals that the medicinal attributes of paragis have been utilized for many reasons. Investigations have shown that paragis contains bioactive components known to possess antimicrobial activity. The present study conducted an experiment to determine the antifungal ~~potential~~ (FP-Gen) of paragis leaf extracts against Colletotrichum species. This study made use of agar block method in isolation of the test organism in the prepared Potato Dextrose Agar media. Paper disc diffusion assay was done using crude ethanolic leaf extract in different concentrations and extraction techniques, Kupchan Partitioning Method (Polar, Semi-polar, Non-polar fractions), and ethanolic crude in boiled, and not boiled treatments in determining its antifungal ~~potential~~ (FP-Gen). All experiments were performed in triplicate and were presented as mean + SD. Statistical analysis performed using One-way, Univariate ANOVA, and Tukey's HSD Multiple Comparison Test. The findings of this study showed and suggests (EC) that the extraction efficiency favors the highly polar solvents than the non-polar solvents in determining the antifungal ~~potential~~ (FP-Gen) of E.

indica against Colletotrichum spp.. The findings of this study attested that this specific type of grass could (EC) potentially (EC) be used as biofungicide agent.

**B15 · 131 words · 0 hedges · 0.0 per 100 words**

The study investigates bamboo-aloe vera composite plates as eco-friendly alternatives to non-biodegradable materials like styrofoam and silver foil paper plates. Researchers conducted experiments on plates made from bamboo leaves with and without aloe vera coating, assessing properties like biodegradability, tensile strength, water uptake, and melting temperature. The biodegradable plate developed exhibited the following properties: Bamboo fiber plates (T1) and bamboo fiber plates coated with aloe vera (T2) degrade faster than styrofoam and silver foil plates; Bamboo fiber plates coated with aloe vera (T2) showcased the highest tensile strength compared to other treatments Bamboo fiber plates (T1) and bamboo fiber plates coated with aloe vera (T2) uptake more water than styrofoam and silver foil paper plates; The significant differences in melting temperature ~~indicate~~ (FP-Gen) variations in thermal stability among the different plate materials.

**B16 · 223 words · 5 hedges · 2.24 per 100 words**

Methicillin-resistant Staphylococcus aureus (MRSA) is a common and potentially (EC) severe pathogen that has developed resistance to several types of antibiotics. MRSA bacteria ~~can~~ (FP-Cap) survive long on surfaces and objects, including second-hand clothing (SHC). The increasing popularity of SHC or “ukay-ukay” in the Philippines, coupled with potential (EC) public health concerns associated with transmitting pathogenic bacteria, emphasized this research’s need. Thus, this study detected the presence of S. aureus and

MRSA in second-hand clothing. A total of fifteen most frequently purchased second-hand clothing (i.e., blouses, tee shirts, crop tops, tank tops, and hoodie jackets) were collected from [REDACTED]. Bacteriological analyses demonstrated the presence of Staphylococcus species across all samples, with growth in high-salt-containing culture broth and characteristic growth on Mannitol Salt Agar (MSA). Biochemical tests revealed positive catalase and Gram-positive cocci in clusters typical of S. aureus. Antibiotic susceptibility testing indicated the presence of both methicillin-sensitive S. aureus (MSSA) and methicillin-resistant S. aureus (MRSA) strains among the collected samples, highlighting the potential (EC) public health risks associated with the handling and purchasing of second-hand clothing. This study, therefore, underscores the need for proper precautions to mitigate the spread of potentially (EC) harmful bacteria, including MRSA, on second-hand clothing. It highlights the importance of raising awareness among both vendors and consumers about the potential (EC) health risks associated with these garments.

**B17 · 243 words · 2 hedges · 0.82 per 100 words**

The increasing prevalence of antibiotic resistance among bacterial pathogens, particularly Staphylococcus aureus and Escherichia coli, necessitates the search for alternative antimicrobial agents. Medicinal plants have been widely explored for their bioactive compounds with antibacterial and antioxidant properties. However, there is a paucity of research on the phytochemical composition and biological activities of Ptychosperma propinquum (Macarthur Palm) fruit. This study aimed to assess the phytochemical composition, antioxidant ~~potential~~ (FP-Gen), and antibacterial activity of P. propinquum

fruit ethanolic extract using Thin Layer Chromatography (TLC)-based techniques. The fruit extract was subjected to TLC phytochemical screening using various spray reagents to detect alkaloids, glycosidic flavonoids, steroids, anthraquinones, anthrones, coumarins, higher alcohols, essential oils, phenols, tannins, and flavonoids. Antioxidant activity was evaluated using the TLC-DPPH (2,2-diphenyl-1-picrylhydrazyl) assay, while antibacterial activity was assessed using the TLC-MTT ((3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide) overlay assay against S. aureus (ATCC 25923) and E. coli (ATCC 25922). Results revealed the presence of phenolic compounds, flavonoids, steroids, and alkaloids in the fruit extract. The TLC-DPPH assay demonstrated notable antioxidant activity, as ~~indicated~~ (FP-Gen) by the reduction of DPPH radicals. The TLC-MTT overlay assay showed that the extract exhibits antibacterial activity against S. aureus but was ineffective against E. coli, suggesting (EC) selective antibacterial ~~potential~~ (FP-Gen). The findings indicate that P. propinquum fruit ethanolic extract contains bioactive compounds with antioxidant and selective antibacterial properties. Further studies ~~should~~ (FP-Rec) focus on the isolation and characterization of its active compounds, their mechanisms of action, and their potential (EC) pharmaceutical applications.

**B18 · 286 words · 1 hedges · 0.35 per 100 words**

This study investigates mercury detection in water and green mussels and oysters in Abucay, Bataan, and its economic effects on local fishers. The research evaluates water quality by analyzing chemicals characteristics in water such as pH, Biological Oxygen Demand (BOD), and Chemical Oxygen Demand (COD) at upstream, midstream, and downstream locations. Findings

revealed that mercury levels in water were in the result from physio- chemical analysis of water revealed pH levels of 7.2 upstream and midstream and 7.1 downstream, all within the acceptable range of 6.5 – 8.5 set by DENR. BOD was consistently 2 mg/L across all sites COD was recorded at 4.0mg/L upstream and midstream but increased to 7.1 mg/L downstream. However, the mercury detection in water from upstream to downstream is <0.0001 mg/L which below the DENR regulatory limit of 0.0002 mg/L, and mercury in shellfish were in the result is not detected or “ND” staying within the FAO safety threshold of 0.5 mg/kg. These results confirm that seafood from the study area remains safe for consumption. However, despite the absence of contamination, economic challenges persist. In the p-value in the analysis of economic impact is 0.064 and p-value of fishing practices is 0.069 are slightly above 0.05, the result suggests (EC) that there is no strong statistical evidence that heavy metal contamination alone is significantly affecting fishers’ income and practices. The study highlights the need for continued water quality monitoring, stricter pollution control measures, and increased institutional support to address economic threats. Strengthening public mindfulness campaigns and applying sustainable fishing practices are also essential to maintaining both seafood safety and the livelihoods of littoral communities. The findings contribute to policy development and environmental operation, assuring a balance between profitable stability and marine ecosystem protection.

**B19 · 290 words · 1 hedges · 0.34 per 100 words**

Wetlands are becoming popular as important tourist destinations due to their unique natural beauty and diverse wildlife. These vital ecosystems are at risk from the unregulated and unsustainable tourist visits that cause detrimental

effects on the environment and host communities. Hence, sustainable planning and management approaches ~~should~~ (FP-Rec) be considered. This study used Boullon's Carrying Capacity Mathematical Model adopted from the Ecosystems Research and Development Bureau for computing the Tourism Carrying Capacity (TCC) of Sasmuan Pampanga Coastal Wetlands. Data on the parameters needed in TCC were collected through Key Informant Interviews, direct observation of the site, and a questionnaire-based survey for the tourism-related preferences and residents' perceived impact of tourism activities. The Real Carrying Capacity per day was found to be a total of 13 boats and 156 visitors for fishing activity on the river stretch of [REDACTED], 14 visitors in the [REDACTED], while 420 visitors for aquaculture farm fishing I and 317 visitors in aquaculture farm fishing II, while 8 visitors for bangus feeding I and a total of 9 visitors for bangus feeding II. Only 5 visitors were allowed for birdwatching and 65 visitors for swimming activity. The results revealed that present visitations in these activities are still below the carrying capacity. For descriptive statistics, three-hundred fifty (350) respondents from five coastal barangays of SPCW were purposively selected. Results showed that respondents perceived tourism activities have a positive impact and equal perceptions in ecological (62.00%), economic (59.71%), social (59.71%), and visitor experiences aspects (50.86%). However, tourism activities in the area are perceived to have a negative impact on SPCW's local facilities (54.00%). Thus, local government ~~should~~ (FP-Rec) address the possible (EC) negative impacts of tourism activities and manage the carrying capacity for sustainable tourism.

**B20 · 223 words · 1 hedges · 0.45 per 100 words**

Many farmers have been facing many problems towards their crops. One of the common factors affecting the production of eggplant is the pathogenic diseases. The fungal disease commonly found in Eggplant (Solanum melongena) is Cercospora leaf spot, a plant-pathogenic fungus, Cercospora melongenae. Therefore, this study aims to examine the ~~potential~~ (FP-Gen) of Trichoderma harzianum, an endophytic fungus, as a biocontrol agent for Cercospora leaf spot-on eggplants in the Philippines. Cercospora melongenae is a plant-pathogenic fungus that causes significant damage to eggplant crops, and the overuse of chemical pesticides has adverse effects on human health and the environment. As such, using alternative approaches, such as endophytic fungi, has gained attention from researchers. The study used thirty isolated leaf samples, twenty-one subcultures has been done and four different pathogenic fungi were identified. Moreover, there is an antagonistic effect of Trichoderma hazianum against the pathogenic fungi that cause disease in eggplant crops. Hence, the study aimed to investigate the antagonistic ~~potential~~ (FP-Gen) of Trichoderma harzianum among the four isolates. The results ~~indicate~~ (FP-Gen) that the mean effect of Trichoderma harzianum on the test variables varied across the three days of observation. However, there was no significant difference in the effect of Trichoderma harzianum among the test variables on any of the three days, suggesting (EC) that the fungus had a similar impact on the growth of all four test variables.

## Veterinary Medicine (VM1–VM20)

### VM1 · 211 words · 0 hedges · 0.0 per 100 words

This study was [C] conducted to determine the anesthetic effect of Zoletil® in zebrafish (Danio rerio). A total of 25 zebrafish with similar body sizes were used for three treatments. Treatment 1 was Zoletil® waterbath. The dosage started at 10mg/L and it continued to increase by 10mg/L with a maximum of 50 mg/L. Stage 1 of anesthesia was observed in T1, but it did not progress to Stage 2 and Stage 3 of anesthesia. Treatment 2 was waterbath mixed with Sodium bicarbonate. The dosage that was used started at 20 g/L. In dosages 20 g/L and 30 g/L, Stage 1 and Stage 2 anesthesia was observed, but it failed to induce Stage 3. Whereas dosages 40 g/L, 50 g/L, and 60 g/L were able to induce all three stages of anesthesia, Treatment 3 was a combination of Zoletil® and sodium bicarbonate. In dosage of 5 mg/L of Zoletil® and 30 g/L of sodium bicarbonate, Stage 3 was observed. T1 revealed that Zoletil® alone is not enough to fully induce Stage 3 anesthesia in zebrafish. However, when sodium bicarbonate was added, Stage 3 was achieved, while using only a small amount of Zoletil®. Additionally, it was also revealed that 40 g/L of sodium bicarbonate was sufficient to induce Stage 3 anesthesia in zebrafish.

### VM2 · 358 words · 3 hedges · 0.84 per 100 words

This study determined the knowledge, attitude, and practices of PSAU students regarding stray animals. A total of 386 respondents were calculated using Slovin’s formula. The respondents were reached through social media platforms such as facebook, messenger, and through text messages. The data collected was entered into an Excel database and was subjected to a Chi-square

test to examine associations with the socio-demographics of the respondents; a p-value of 0.5 was considered significant. Descriptive analysis was done to examine relationships between categorical variables, particularly focusing on the influence of socio-demographic profile differences. The results in overall knowledge of PSAU students regarding stray animals showed that over half (56%) of the students were aware and have moderate knowledge about stray animals, Moreover, 32.6% of the students showed good knowledge regarding stray animals, and only 11.4% of them had poor knowledge regarding stray animals. In terms of attitude, 92% of the respondents showed good attitude, indicating (EC) a strong sense of empathy and concern for the welfare of stray animals on campus. A smaller portion of students, 7.3%, expressed a moderate attitude, and only 0.8% of the respondents showed a poor attitude towards stray animals. Among the respondents, only 19.7% showed a good level of practice, and 39.1% of the respondents showed a moderate level of practice and 41.2 % shows poor level of practice. In terms of association of demographic profile with the KAP, the findings showed that sex, year level, and pet ownership had a significant effect on students' knowledge and practices toward stray animals. A strong relationship was found between sex and knowledge ($p = 0.000$), with female students more likely (EC) to have moderate to good knowledge. Year level also influenced knowledge ($p = 0.010$), with first-year students showing poor knowledge and higher-year students demonstrating good knowledge. For practices, sex ($p = 0.005$), year level ($p = 0.010$), and pet ownership ($p = 0.044$) all showed significant relationship. Female students, higher-year students, and those who owned pets were more likely (EC) to engage in responsible and humane behavior toward stray animals. The results

obtained served as baseline information for future endeavors in promoting responsible pet ownership for the welfare of companion animals and the community.

**VM3 · 296 words · 0 hedges · 0.0 per 100 words**

This study assessed the profiling, occurrence and antibiotic resistance of Escherichia coli and Salmonella spp., in retail chicken eggs sold in [REDACTED]. The study aimed to profile handling practices of egg vendors, detect E. coli and Salmonella spp. on retail chicken eggs sold in [REDACTED] and assess the degree of antibiotic susceptibility of the bacterial isolates. A total of 16 vendors were evaluated on their egg handling practices, evaluated using a survey checklist, focusing on the assessment of demographic profile and their egg handling practices aligned with the Code of Hygienic Practice for Eggs and Egg Products. For the detection of E. coli and Salmonella spp. isolates, a total of 36 samples were collected, comprised of 16 eggshell surface samples and 16 egg content sample. Samples were subjected to selective culture media, biochemical tests and antibiotic sensitivity testing of bacterial isolates. None of the samples tested positive for Salmonella spp. However, 19% of egg samples tested positive for E. coli on eosin methylene blue (EMB) agar, characterized by green metallic sheen colonies. The 6% of positive samples were detected on the egg shell surface, while 13% of isolated E. coli were located on the egg's homogenized yolk and albumen. Antibiotic sensitivity testing using disk diffusion method following CLSI guidelines revealed 100% resistance to chloramphenicol, tetracycline, and streptomycin. While 33% showed [C] resistance to meropenem. On the other hand, the evaluation of retail egg vendors showed a major issue of reusing the egg trays across vendors which has been strongly

linked to bacterial growth on eggshells, yolks and albumen, with p-values of 0.030 and 0.014, respectively. The findings ~~suggests~~ (FP-Rec) the need for assessment of knowledge, attitudes and practices of retail egg vendors about food safety and regular antimicrobial resistance monitoring on retail eggs.

**VM4 · 250 words · 0 hedges · 0.0 per 100 words**

The Escherichia coli and Campylobacter jejuni are two of the foodborne pathogens affecting public health around the world. A cross-sectional bacteriological study was conducted from January to February 2025 in [REDACTED]. This study presents the bacteriological analysis of E. coli and C. jejuni from 12 raw buffalo milk samples from 12 vendors. Data were gathered from vendors through modified and patterned questionnaires to assess demographics and hygiene practices of vendors. Results indicated that most vendors lacked formal training in milk-handling (100 %) did not use refrigeration, new bottles, or any preservation methods (100%). The majority of the vendors that participated in the study sell milk at public markets (66.7 %), while others sell it through peddling (25 %) or by using vehicles (8.3 %). The detection and isolation of these pathogens was carried out by using selective culture media and biochemical tests for further identification. The C. jejuni was not detected in any of the samples (0%) on Charcoal Cefoperozone Deoxycholate agar (CCDA). In contrast, all 12 samples (100%) tested positive for E.coli on MacConkey agar, characterized by moist and pink to red colonies, on Eosin Methylene Blue agar, characterized by its green metallic sheen and nutrient agar, characterized by opaque, creamy/off-white colonies. Isolates were subjected to biochemical tests, such as Triple Sugar Iron (TSI), IMViC series, Catalase, Hippurate hydrolysis, and Gram staining. These findings

~~suggest~~ (FP-Rec) the need to improve training, regulation of milk storage, and the enforcement of labeling and sanitation standards to improve public health safety in the milk supply chain.

**VM5 · 256 words · 1 hedges · 0.39 per 100 words**

The prevalence of dermatophytosis and emerging drug resistance underscores the need for alternative therapeutic approaches, leading to a focus on medicinal plants. This study explores the ~~potential~~ (FP-Gen) antifungal properties of Muntingia calabura L. methanolic leaf extracts against Trichophyton mentagrophytes, a dermatophyte species. The methanolic leaf extract of M. calabura L. exhibited various phytochemicals, including anthraquinones, anthrones, alkaloids, glycosides, phenols, tannins, flavonoids, saponins, higher alcohols, and essential oils, presenting a promising natural resource for antifungal treatment. Agar well diffusion, broth dilution, and determination of minimum fungicidal concentration, were employed to evaluate the efficacy of the extract. Statistical analysis revealed a significant difference among the treatments used, with Ketoconazole (Treatment 2) as a positive control displaying high susceptibility (42.5667 mm). Treatment 3, containing 200 mg/mL M. calabura L. methanolic leaf extract, showed that the microorganism utilized was susceptible to the leaf extract at this concentration as it exhibited a DIZ of 26.4667 mm. Treatment 4 containing 100 mg/mL M. calabura L. methanolic leaf extract showed intermediate susceptibility as the DIZ was measured at 15.3667 mm. Treatment 5, composed of 10 mg/mL M. calabura L. methanolic leaf extract, displayed an inhibition zone of 9.70 mm. Treatment 1, which contained distilled water, did not display any inhibition against T. mentagrophytes. The minimum inhibitory concentration (MIC) of M. calabura

L. methanolic leaf extract was determined at 10 mg/mL, consistent with the minimum fungicidal concentration. These findings highlight the potential (EC) of M. calabura L. as a source of antifungal agents, warranting further investigation into its therapeutic applications and formulation for dermatophytosis management.

**VM6 · 278 words · 0 hedges · 0.0 per 100 words**

This study determined the hematological and electrolyte values of island-born New Zealand White (NZW) rabbits (Oryctolagus cuniculus). Eight (8) adult rabbits (4 female and 4 male) with initial weight between 2.5 to 2.7kgs were used for sample collection. Around 3.0 ml and 0.5 ml of blood was collected from the central auricular artery and the lateral saphenous vein, respectively of each rabbit. Hematological tests for packed cell volume (PCV), red blood cells (RBC), hemoglobin (HGB), mean corpuscular hemoglobin (MCH), mean corpuscular hemoglobin concentration (MCHC), mean corpuscular volume (MCV), white blood cells (WBC), neutrophils, eosinophils, lymphocytes, monocytes, and serum electrolyte tests for sodium (Na), potassium (K), magnesium (Mg), and calcium (Ca) were requested from Hi-Precision Diagnostics Laboratory. Results were gathered and mean values were determined using MS Excel and subjected to statistical analysis using t-test through statistical package for the social sciences (SPSS). The mean values of each hematological parameter of male and female rabbits, respectively were: PCV (37.32% and 37.4%); RBC (5.91x106/mm3 and 5.76x106/mm3); HGB (12.32g/dL and 12.37g/dL); MCH (20.63pg and 21.47pg); MCHC (33.03% and 32.90%); MCV (62.57fL and 64.52fL); WBC (7.39x109/L and 8.17x109/L); neutrophils (43.30% and 51.85%); eosinophils (0.10% and 0.10%); lymphocytes (52.47% and 41.22%); monocytes

(2.18% and 4.61%); and basophils (1.94% and 2.21%). The mean values of each electrolyte parameter of male and female rabbits, respectively were: sodium (140.25mmol/L and 143.25mmol/L); potassium (4.32mmol/L and 4.37mmol/L); magnesium (0.87mmol/L and 0.98mmol/L); and calcium (3.41mmol/L and 3.21mmol/L). Mean values for PCV, RBC, HGB, MCH, MCHC, MCV, WBC, neutrophils, eosinophils, lymphocytes, monocytes, basophils, potassium, magnesium, and calcium were not significantly different while mean values for sodium between male and female NZW rabbits were found to be statistically different ($p<0.05$).

**VM7 · 174 words · 1 hedges · 0.57 per 100 words**

Canine heartworm (Dirofilaria immitis) infection is a significant health concern for dogs in many regions. This study investigated the association between clinical signs and microscopy/serological detection of heartworm at the [REDACTED]. Canines below six months of age and those receiving annual macrocyclic lactones were excluded. A questionnaire was utilized to collect data, and 100 single step rapid test kits canine heartworm Ag (AccuVet®) were employed along with direct blood smears. However, no positive results were observed in the diagnostic tests conducted. Thus, no significant association between clinical signs and heartworm infection could (EC) be established. Recommendations include increased awareness and education about heartworm prevention, regular testing, evaluation of other diagnostic methods, collaboration and research, continuous monitoring of heartworm prevalence, individualized risk assessment, and collaboration with pet owners. These measures ~~can~~ (FP-Gen) collectively contribute to the effective management and prevention of heartworm disease in the local canine population. Further

research with larger sample sizes and diverse populations is ~~needed~~ (FP-Rec) to validate these findings and enhance our understanding of heartworm infection in the study area.

**VM8 · 253 words · 1 hedges · 0.4 per 100 words**

This study investigated the in vitro anthelmintic efficacy of aqueous leaf extracts from tamarind (Tamarindus indica) and jackfruit (Artocarpus heterophyllus) against Fasciola spp. A total of ten treatment groups were established to evaluate the effects of tamarind leaf extract (TLE) and jackfruit leaf extract (JLE) at various concentrations (5%, 10%, 25%, and 40%) on the motility and viability of adult Fasciola spp., alongside positive (Albendazole) and negative (Nutrient broth) controls. Four adult flukes were placed in each experimental plate, with each treatment tested in triplicate. Motility was monitored every 10 minutes, and histopathological examinations were conducted to assess morphological changes in the tegument of the flukes. Phytochemical screening confirmed the presence of alkaloids, flavonoids, glycosides, phenols, quinones, saponins, and tannins in both TLE and JLE. The results demonstrated that both extracts possess fasciolocidal activity, with a dose-dependent reduction in motility and increased mortality of the flukes. TLE exhibited a more potent effect, achieving complete immobilization at 40% concentration within 7.84 minutes and lower $LD_{50}$ (2.680 mL/mL) and $LD_{90}$ (3.209 mL/mL) values compared to JLE ($LD_{50}$ = 5.156 mL/mL; $LD_{90}$ = 6.165 mL/mL). Nevertheless, JLE also caused significant reductions in motility and tegumental damage at higher concentrations. Histopathological analysis confirmed that both extracts induced severe alterations in the fluke's tegument, including vacuolations, separation of layers, and disintegration of

spines and tegumental swelling. These findings confirm the potential (EC) of both plant extracts as natural anthelmintics, with tamarind showing superior efficacy. The study supports the development of plant-based alternatives for the sustainable control of fasciolosis in livestock.

**VM9 · 430 words · 1 hedges · 0.23 per 100 words**

Plastic pollution has been a long-standing environmental concern due to improper waste management. Its persistence in marine environments, particularly in the form of microplastics, posed potential (EC) health risks to aquatic species and humans through seafood consumption. This study was conducted to detect microplastics in Whiteleg shrimp (Litopenaeus vannamei) sold at the [REDACTED]. A total of 10 shrimp samples per stall were collected, equating to 30 shrimps overall from the 3 stalls available. Composite sampling was applied in this study; hence, all individual shrimps from each stall were counted as one sample. The samples were separated into two treatments: Treatment 1, which consisted of the hepatopancreas and gastrointestinal tract of the shrimp, and Treatment 2, which consisted of the shrimp muscle. The samples were subjected to light microscopy and were categorized as follows: Treatment 1 Stall 1 (T1S1), Treatment 1 Stall 2 (T1S2), Treatment 1 Stall 3 (T1S3), Treatment 2 Stall 1 (T2S1), Treatment 2 Stall 2 (T2S2), and Treatment 2 Stall 3 (T2S3). Each sample was added to a 10% KOH solution for the digestion of soft tissue. Afterwards, 60% sodium iodide was used for the density separation of microplastics and undigested organic matter. The supernatant was collected using a pipette and then transferred to a slide before subjecting to a light microscope. In each slide, 10 µl of sample was placed. This was repeated five times in separate slides. A total of 181 microplastics were detected across all

samples, wherein the concentration of microplastics was higher in Treatment 1 (74.03%). Fragments (39.02%) were the most dominant shape, followed by fibers (23.20%), lines (17.13%), and then pellets (3.31%). The most dominant color was gray (33.15%), followed by blue (24.31%), black (11.6%), red (9.39%), clear (7.73%), brown (6.63%), violet (3.31%), yellow (3.31%), and green (0.55%). Most of the particles ranged in size from 11–50 µm (58.01%), followed by <10.9 µm (29.83%), 51–100 µm (6.08%), 101–200 µm (5.52%), and >201 µm (1.10%). Through random sampling, one microplastic was selected from each shape for polymer analysis using Fourier Transform Infrared Spectroscopy (FTIR). The fragment had a 97% match with high-density polyethylene, the fiber had a 90% match with an olefin copolymer, and the line sample had a 96% match with polypropylene, while the pellet sample was rejected for analysis due to its small size. To reduce microplastic pollution, the findings ~~recommend~~ (FP-Rec) removing the shrimp's GIT during preparation to avoid ingestion, use fishing gear made from durable or natural materials, regularly maintain and replace worn equipment, and properly dispose of plastics instead of discarding them into water bodies.

**VM10 · 277 words · 0 hedges · 0.0 per 100 words**

This study assessed the existence and antibiotic resistance characteristic of Escherichia coli from poultry farms in Central Luzon using Regional Animal Disease Diagnostic Laboratory (RADDL) III data for 2022 to 2023. This study targeted the profiling of Escherichia coli bacteria isolates according to year, province, and type of sample. Also, it checked susceptibility to different antibiotics of Escherichia coli-positive poultry samples. The association of the antibiotic susceptibility tests with years and provinces was also calculated. The

study results showed that the susceptibility of heart samples was ended with 100% to amikacin, ceftazidime, cefotaxime, chloramphenicol, ciprofloxacin, gentamicin, imipenem, and meropenem. On the other hand, reduced susceptibility was observed for trimethoprim-sulfamethoxazole and tetracycline 50% in each. In general, lung samples were [G] sensitive to all antibiotics except amoxicillin-clavulanic acid. However, liver samples were 100% susceptible to amikacin, cefotaxime, ceftazidime, ceftiofur, chloramphenicol gentamicin, imipenem, and meropenem. Nasal swabs were 100% susceptible to ceftiofur and florfenicol, with a lower susceptibility for amoxicillin-clavulanic acid at 50% and then amoxicillin only 33%. Tracheal swabs showed 100% susceptibility for amikacin, ceftazidime, cefotaxime chloramphenicol, gentamicin, imipenem, and meropenem. Nonetheless, they only were 50% sensitive to trimethoprim-sulfamethoxazole and tetracycline. Chick samples had different susceptibility meropenem, experienced as the most susceptible antibiotic (87%) and sulfamethoxazole / trimethoprim, while sulfamethoxazole was 25%. In contrast, the data showed that lung and chick samples exhibited marked resistance. In the lung samples, resistance was observed to all tested antibiotics, and only a few were susceptible to them, while chick specimens with high resistances were found against ceftazidime, chloramphenicol, ciprofloxacin, and tetracycline. Also, nasal swabs were highly resistant to multiple antibiotics. The lung samples were resistant to all antibiotics, apart from meropenem and streptomycin.

**VM11 · 236 words · 0 hedges · 0.0 per 100 words**

This study was conducted to help determine the overall health of stray cats within the vicinity of Dr. Andres M. Luciano [REDACTED], based on their

hematological profiles. All 16 cats that were captured, regardless of age and sex, were included in this study. The stray cats' hematological profiles were compared to normal parameters to assess their health status and whether they had the ~~possibility~~ (FP-Gen) of any underlying diseases based on the blood profiling. The results were also used to determine the stray cat's eligibility to undergo surgery for the trap, neuter, vaccinate, and release (TNVR) program. Blood profiling results showed that red blood cell indices such as RBC count, hemoglobin, hematocrit, MCV and MCHC all fell within normal ranges, however MCH was notably below normal ranges (mean=16.81; reference = 25.8-34), platelet (PLT) count was markedly below the normal range, with a mean value of 75.8 compared to the normal reference range of 300 to 800. White blood cells, lymphocytes, and monocytes were within normal ranges. However, granulocytes exceeded the normal range (mean=9.71; reference =2-8), and the percentages of lymphocytes, monocytes, and granulocytes all remained within normal limits. Blood urea nitrogen and creatinine were within normal limits; however, ALT and AST were both above normal (ALT mean = 105.88, reference= 20 to 100, AST mean 46.96; reference 7-38). All captured cats were deemed eligible for the trap, neuter, vaccinate, and release (TNVR) program.

**VM12 · 182 words · 1 hedges · 0.55 per 100 words**

The African swine fever (ASF) is an ongoing global disease of swine presenting clinical signs similar to hog cholera and causing significant socioeconomic problems. Studies on the detection of ASF virus focus on using viremic blood and lymph tissues from infected pigs at the farm level. Hence, this study aimed to detect ASF virus from suspected pork samples from [REDACTED] using a

survey questionnaire and molecular diagnostic methods. Profiling of meat retail vendors revealed that commercial farms and public markets were their principal sources of meat. PCR assay showed that no sample tested positive for ASF. The result of collective community and government efforts might (EC) be effective in preventing, controlling, and managing the spread of ASF, but it is not conclusive that ASF is not present in the municipality of [REDACTED]. Nevertheless, it is ~~recommended~~ (FP-Rec) to consider monitoring and surveillance of ASF at the market level, and maintain and continue the strict implementation of collaborative efforts to effectively and efficiently prevent and control the further entry and spread of ASF from farm to farm.

**VM13 · 215 words · 0 hedges · 0.0 per 100 words**

Humans and animals have historically utilized immunoglobulin Y (IgY), the most common kind of immunoglobulin generated by B cells in avian species, as a therapeutic treatment for passive immunity. This study was conducted to determine the immunogenicity of freeze-dried egg yolk against canine parvoviral enteritis. Primodog® (Merial) is a monovalent, attenuated canine parvovirus vaccine administered intramuscularly to three white Leghorn chickens. For a total of five weeks of immunization, 0.5 ml of the initial dose per bird and four further doses—each lowered to 0.25 ml—were given at weekly intervals. As described in the Hodek et al. (2013) protocol, a two-step, labor- and cost-efficient process that involves extracting yolks using tap water then precipitating specific IgY at pH 4 until the final concentration reaches 8.8% with sodium chloride was used to isolate IgY from the hyperimmunized egg yolks. Isolated IgY were freeze-dried and the products' immunogenicity was assessed using sandwich ELISA testing. The data gathered demonstrated the

concentration of IgY specific to the canine parvovirus (CPV) following five weeks of immunization and freeze drying. The first week of immunization had the highest measured titer with a mean of 0.796. One way ANOVA was used and the differences between the groups were statistically significant. There were variations in the mean antibody titer measured among samples.

**VM14 · 183 words · 0 hedges · 0.0 per 100 words**

Companion animals play vital roles in enhancing human well-being but remain vulnerable to a range of toxicants including household products, medications, food items, and ornamental plants. This study assessed the level of awareness among students of Pampanga State Agricultural University (PSAU) regarding common toxicants affecting companion animals. Using a cross-sectional, descriptive design, a structured questionnaire was distributed to a stratified random sample of 393 students. The instrument focused on awareness of specific toxicants, personal experiences, and sources of information. Findings revealed that while chocolate (87.28%) and alcoholic beverages (59.79%) were widely recognized as toxic, awareness of plant-based toxicants such as Philippine lily (18.06%) and avocado (23.41%) was limited. Majority of respondents (42.5%) demonstrated only slight awareness, and 65.14% reported previous unintentional exposure of pets to toxicants—most commonly garlic. Chi-square analysis revealed significant associations between course of study, pet ownership, and source of information with awareness levels. Multiple regression analysis ~~indicated~~ (FP-Gen) that knowledge significantly predicted self-assessed awareness, while personal experience did not. The findings highlight critical knowledge gaps and

underscore the need for targeted educational interventions using accessible media such as social platforms.

**VM15 · 258 words · 1 hedges · 0.39 per 100 words**

The study assessed the anti-cancer and anti-angiogenic properties of the different varieties of mulberry (Morus alba L.), namely, var. S54, var. Alfonso and var. Batac leaves crude extract. It also determined the lethal dosage (LD50) and the percent reduction in the mean number of blood vessel branching points in the eggs treated with different Morus alba L. leaf extracts. The anti-cancer ~~potential~~ (FP-Gen) of the extract was determined using a brine shrimp lethality assay, and after three and six hours of exposure, the surviving brine shrimps were counted. The brine shrimp lethality (BSL) assay results revealed that the degree of brine lethality was directly proportional to the concentration of the different varieties (var. S54, Alfonso, and Batac) of leaf crude extract. Maximum mortalities occurred at a concentration of 100,000 parts per million (ppm), whereas the least mortalities were at ten ppm. The lethality of the leaf extracts used to brine shrimps proved the plant's anti-cancer activity. Moreover, a chorioallantoic membrane (CAM) assay was used to test the anti-angiogenic property. The assay results revealed positive angiogenic inhibitory effects when the blood vessels formed in the eggs treated with leaf crude extract of S54, Alfonso and Batac were reduced by 74.63%, 68.49%, and 68.02 %, respectively, compared to the control group. Furthermore, the result of the study suggests (EC) that var. S54 presented the most potent anti-cancer and anti-angiogenic activity compared to the other varieties (Alfonso and Batac). It had the highest mortality rate after 3 and 6 hours of exposure to the treatment and the highest recorded percentage reduction (%) of 74.63.

**VM16 · 261 words · 0 hedges · 0.0 per 100 words**

Noni (Morinda citrifolia) is recognized to have a wide range of therapeutic use including illnesses like inflammation, back pain, diabetes, hypertension, cancer, joint problems, and skin allergies. Adding to these arrays of claims, this study was conducted to determine the wound healing efficacy of Noni fruit ethanolic extract and its antimicrobial activity against Staphylococcus aureus strain ATCC 29213. The agar well diffusion method was done to determine the concentration of Noni ethanolic extract yielding the highest zone of inhibition. The treatment were as follows: Treatment 1 (T1) with 40% concentration Noni extract; Treatment 2 (T2) with 20% concentration noni extract; Treatment 3 (T3) with 10% concentration Noni extract; a positive control with 2% Mupirocin; and a negative control with pure petroleum jelly. Results revealed that Treatment 3, with 40% concentration of Noni had the highest zone of inhibition in vitro with comparable antibacterial activity as the positive control against S. aureus. The 40% Noni concentration was used in the in vivo part of the experiment. A total of nine male albino mice were used to test the antibacterial activity of three different treatments given once a day for seven days. Treatment 1 was the 40% concentration of Noni ethanolic extract, T2 was the positive control with 2% Mupirocin (Mupisaph®) and T3 was the negative control with pure petroleum jelly. The results showed that applying 40% Noni ethanolic extract and 2% Mupirocin ointment exhibited a faster wound closure in vivo though not significantly different. With the in vitro and in vivo results, Noni fruit ethanolic extract with 40% concentration has [G] antibacterial property.

**VM17 · 232 words · 1 hedges · 0.43 per 100 words**

This study examined the effect of utilizing butterfly pea flower aqueous extract (BPFAE) on the survival rate and locomotor activity of zebrafish larvae, as well as the water quality parameters. A total of 360 five-day post-fertilization zebrafish larvae were randomly distributed into four treatments— 0 parts per thousand (ppt) BPFAE, 5 ppt BPFAE, 10 ppt BPFAE, and 15 ppt BPFAE, with three replicates each, following a completely randomized design. The butterfly pea flower aqueous extract was administered to test subjects for 17 days. The results were analyzed using one-way analysis of variance (One-way ANOVA), $P<0.05$. Significant comparisons were subjected to Tukey HSD. The results revealed that 10 ppt BPFAE is the optimal inclusion level among all the treatments. This concentration resulted in the highest survival rate (84.44%) as compared to the other treatment groups — 0 ppt (65.55%), 5 ppt (62.22%), and 15 ppt (68.88%). Meanwhile, no significant effects on the locomotor activities of zebrafish larvae were observed in any of the treatments. However, notable reductions in water pH during the first week of administration of treatments, and suboptimal dissolved oxygen concentrations were observed. Additionally, minimal temperature changes within the optimal range were recorded. Lastly, the levels of ammonia, nitrite, and nitrate remained at zero throughout the duration of the experiment. The results demonstrated the potential (EC) of butterfly pea flower aqueous extract in improving the survival of zebrafish at its larval stage.

**VM18 · 267 words · 0 hedges · 0.0 per 100 words**

Zoonotic diseases are a significant public health concern, particularly those that ~~can~~ (FP-Cap) be transmitted from dogs to humans. This study determined the level of awareness among dog owners regarding canine zoonotic diseases.

Specifically, it (1) determined the participants' sociodemographic profile, (2) identified their level of awareness on selected zoonotic disease such as rabies, leptospirosis, ancylostomiasis and toxocariasis, and dermatophytosis, (3) identified the primary sources of information about zoonotic diseases, and (4) determined whether there was a significant association between sociodemographic variables and awareness on zoonoses. Data gathered from a total of 86 dog owners visiting the selected veterinary clinics in [REDACTED], was subjected to descriptive statistics (frequency distribution and weighted mean), while Pearson's correlation determined association between awareness on zoonotic diseases and the sociodemographic factors. Results showed that participants were mostly 26 to 40 years old (43%), females (66.3%), college graduates (67.4%), from lower middle-income families (54.7%), had owned dogs for more than five years (46.5%), and were owners of at least one to two dogs (44.2%). Participants exhibited an overall moderate level of awareness (mean=3.30) on canine zoonotic diseases. The dog owners were very aware of rabies (mean=3.74) and leptospirosis (mean=3.73); while moderate awareness was found on toxocariasis and ancylostomiasis (mean=3.01), scabies (mean=3.05), and dermatophytosis (mean=2.96), The internet (38.3%) and veterinarians (36%) were the primary sources of information on canine zoonotic diseases. Using Pearson's correlation, significant associations ($p<0.05$) were found between awareness on zoonotic diseases and sociodemographic factors, such as educational attainment (p-value=0.000), monthly income (p-value=0.007), and years of dog ownership (p-value=0.001).

**VM19 · 172 words · 0 hedges · 0.0 per 100 words**

Salmonella spp. and Escherichia coli are two significant bacteria that are a source of human foodborne illness. They result from consuming meat or fish that bacteria have contaminated. The purpose of this study was to determine the anti-microbial resistance of isolated Salmonella spp. and E. coli from commercially available fish in the [REDACTED]. A total of 30 whole fish, three (3) different fish in 10 different stalls were used to test Salmonella spp. and E. coli. Six (6) out of 30 samples in XLD were Salmonella spp. positive, where they exhibited red or yellow colonies with a black in the center, meanwhile three (3) out of 30 samples in EMB were E. coli positive, where they showed a metallic green sheen. In the antibiotic sensitivity testing of six (6) positive samples in Salmonella spp., four (4) showed resistance to ceftazidime (33.33%), chloramphenicol (16.67%), imipenem (16.17%), and tetracycline (83.33%). Of the three (3) positive samples, E. coli exhibited resistance to ceftazidime (33.33%), chloramphenicol (33.33%), imipenem (66.67%), and tetracycline (66.67%).

**VM20 · 271 words · 0 hedges · 0.0 per 100 words**

This study evaluated the knowledge, attitudes, and practices (KAP) of Pandi, Bulacan, Philippines residents regarding rabies, a fatal viral zoonotic disease posing significant public health risks. A cross-sectional survey was conducted among 384 respondents using structured questionnaires to assess awareness, perceptions, and preventive measures against rabies. Statistical analyses, including Chi-square tests and multiple linear regression, were employed to examine associations between socio-demographic factors and KAP scores and predictive relationships among knowledge, attitudes, and practices of

respondents. The results indicated that most respondents demonstrated high knowledge about rabies (69.3%), particularly regarding transmission and prevention, However, misconceptions persisted, notably about rabies' incurability (39.8%) and the role of neutering in disease control (50.3%). Attitudes toward rabies prevention were predominantly positive (77%), as were reported practices (97%). Significant associations were identified between knowledge levels and socio-demographic factors, including educational attainment ($p=0.0008$), where higher education correlated with improved rabies awareness and pet ownership practices ($p=0.035$). For attitudes, significant associations were observed between respondents' age ($p=0.015$) and educational attainment ($p<0.0001$). Practices showed only one significant association with educational attainment ($p=0.016$). Multiple linear regression analysis confirmed interrelationships among KAP dimensions, revealing that higher knowledge scores significantly predicted better preventive behaviors ($p < 0.001$). The findings underscore the necessity of targeted educational campaigns to address misconceptions, enhance community engagement to promote responsible pet ownership, and improve access to veterinary services. Strengthening policies on pet vaccination programs and implementing stricter guidelines for pet care are ~~recommended~~ (FP-Rec) to mitigate rabies transmission risks. These insights align with broader efforts toward rabies elimination in the Philippines by 2030, emphasizing the importance of addressing localized KAP gaps through evidence-based interventions.

## Agriculture, Crop Science (Acs1–Acs10)

### Acs1 · 203 words · 1 hedges · 0.49 per 100 words

This experimental study examined the performance of finger pepper (Capsicum frutescens) treated with fermented banana peel as an organic amendment. The research was conducted in [REDACTED], from March to April 2025. Five treatments were tested: T1 – general recommendation (control), T2 – fermented banana peel applied twice weekly, T3 – 100% recommended fertilizer rate with fermented banana peel (twice weekly), T4 – 75% recommended fertilizer rate with fermented banana peel (twice weekly), and T5 – 50% recommended fertilizer rate with fermented banana peel (twice weekly). Parameters measured included days from transplanting to flowering, plant height at first harvest, number of fruits per plant, weight of fruits per plant (g), number of marketable and non- marketable fruits, and computed yield per hectare (tons/ha). This revealed that the application of fermented banana peel improved all growth and yield parameters assessed. Among all treatments, T5 (50% recommended fertilizer rate with fermented banana peel) yielded the most favorable results in terms of plant height, fruit quantity and weight, and overall productivity. These findings suggest (EC) that fermented banana peel, when applied at an optimal rate in combination with reduced synthetic fertilizer input, ~~can~~ (FP-Cap) significantly enhance the growth and yield performance of finger pepper, offering a sustainable alternative for crop production.

**Acs2 · 184 words · 8 hedges · 4.35 per 100 words**

In this study the researcher used Randomized Complete Block Design with two (2) different treatments which are; Treatment 1- Non-Primed and Treatment 2- Seed Coated with Mycorrhiza. The researcher applied mycorrhiza to enhance root development, drought resistance, and overall growth and yield of mung beans through seed coating. This suggests (EC) that while statistical significance might (EC) not have been observed, the mycorrhiza treatment could (EC) still offer potential (EC) benefits to the plants. The result of the study indicate that there were no statistically significant differences observed across the various measured parameters for the two treatments. However, the researchers still chose to apply mycorrhiza through seed coating to aid in root development, drought resistance, and overall growth and yield improvement in mung beans. This suggests (EC) that while the differences might (EC) not have been significant in the study, the researchers believe that mycorrhiza could (EC) provide beneficial effects in practical agricultural contexts. The study concluded that the application of mycorrhiza positive impact on mung beans during drought and root development highlights its potential (EC) benefits for plant growth. Further research ~~could~~ (FP-Gen) unveil additional insights and applications for optimizing plant health.

**Acs3 · 235 words · 2 hedges · 0.85 per 100 words**

The study was conducted at [REDACTED], from November 2022 to February 2023 to determine the effect of rooting hormones on monstera cuttings. The study used Complete Randomized Design (CRD) with four replications. The pot experiment comprised of main treatments: T1- Control (Water), T2- Aloe Vera gel, T3- Coconut water, T4- Moringa leaves extract, and T5- ANAA. Data were

evaluated using Analysis of Variance (ANOVA). Based on the results of the study, different rooting hormones did not significantly influence the days to shoot emergence, number of shoots, number of leaves, number of adventitious roots, root diameter, length of the longest roots, number of rooted cuttings, and percentage of survival cuttings. Additionally, alternative rooting hormones and synthetic hormone applications had varying survival rates. The highest survival was found in ANAA, followed by control water (distilled or filtered water), coconut water, and aloe vera gel, and the lowest survival was found in moringa leaf extract. In conclusion, the number of rooted cuttings obtained by application of aloe vera gel and moringa leaf extract was comparable in percentage survival with that obtained by application of water (distilled or filtered water) on monstera cuttings. Most can (EC) survive the water treatment application but may (EC) not establish roots and die after several days. However, it ~~can~~ (FP-Cap) be derived that the application of different alternative rooting hormones has a comparable effect with synthetic rooting hormones based on the growth parameters.

**Acs4 · 221 words · 0 hedges · 0.0 per 100 words**

The study was conducted to determine the growth and yield performance of sweet potatoes treated with seaweed extract foliar spray from December 2023 to February 2024 in [REDACTED]. The experiment utilized a Randomized Complete Block Design (RCBD) with five treatments replicated four times. Based on the results of the study, there are significant differences in the length of vines at 60 days after planting (DAP), while other parameters showed no significant differences. These include length of vines at 30 DAP and 70 DAP, number of marketable tubers per plant, number of non-marketable tubers per

plant, length of tubers per plant, weight of tubers per plant, diameter of tubers per plant, computed yield per hectare, and cost and return analysis among the seaweed extract treatments. The study found that sweet potatoes treated with different seaweed extract foliar sprays showed varying levels of growth and yield. According to the Super Bureau's VSP6, all treatments had the highest mean values across all parameters. However, the positive control had the highest mean values for vine length, weight of tubers per plant, and marketable yield per sample area. Therefore, the application of seaweed extract as a bio stimulant did not significantly enhance the growth and yield of sweet potato. However, the negative control ~~can~~ (FP-Cap) reduce production cost and provide the highest return on investment.

**Acs5 · 199 words · 2 hedges · 1.01 per 100 words**

The study on the frequency of fermented azolla application on the growth and yield of onion was conducted at [REDACTED], from December 2024 to April 2025. The experiment consisted of T1 - 75 grams Trichoderma + 1 kg of fermented azolla 1x application, T2 - 75 grams Trichoderma + 1 kg of fermented azolla 2x application, T3 - 75 grams Trichoderma + 1 kg of fermented azolla 3x application, and T4 - 75 grams Trichoderma + 1 kg of fermented azolla 4x application. The parameters used to compare the four treatments were the plant height, number of leaves, bulb diameter, weight of marketable and non – marketable bulbs, and computed yield per hectare. The study found that Trichoderma and Fermented azolla application in onion cultivation significantly improved plant height, number of leaves, bulb diameter, weight of marketable bulbs and non- marketable bulbs, as well as the computed yield per hectare. Among all treatments, T3 (75 grams of Trichoderma + 1kg of

fermented azolla 3x application) was found to be the most effective, yielding the best results. This suggests (EC) that Trichoderma and fermented azolla at proper application frequencies could (EC) be beneficial in enhancing cultivation, promoting both plant growth and economic returns.

**Acs6 · 245 words · 1 hedges · 0.41 per 100 words**

This study was conducted to evaluate the growth performance of Adansonii plant under different soil media over a 90-day period. Implemented in December 2024, the experiment followed a Completely Randomized Design (CRD) with five treatments: T1 — Topsoil only (control), T2 — Topsoil + Rice Hull, T3 — Topsoil + Carbonized Rice Hull, T4 — Topsoil + Vermicompost, and T5 — Topsoil + Biochar. Parameters measured included days to shoot emergence, shoot length, number of leaves, leaf area, root length, number of penetrations, and survival rate. Significant differences were observed across treatments. Treatment 2 (Rice Hull) showed the earliest shoot emergence at 15.29 days, followed by Treatment 4 (Vermicompost) at 15.83 days. These treatments also produced the longest shoots, most leaves, largest leaf areas, and highest survival rates at 87.5%. Treatment 3 (Carbonized Rice Hull) showed moderate growth improvement, while Treatments 1 (Topsoil only) and 5 (Biochar) performed poorly in most parameters, ~~indicating~~ (FP-Gen) limited benefits. Throughout the study, plants in Treatments 2 and 4 exhibited more vigorous and healthier growth, with well-developed roots, expanded foliage, and stable shoot formation. The positive effects of rice hull and vermicompost were attributed to better nutrient availability, improved soil aeration, enhanced microbial activity, and more efficient moisture retention. In conclusion, the use of rice hull and vermicompost significantly enhanced the growth and survival

of Adansonii. These findings highlight their potential (EC) as effective, low-cost, and sustainable soil amendments for ornamental and horticultural cultivation, offering practical options for growers seeking eco-friendly methods to improve plant performance.

**Acs7 · 207 words · 1 hedges · 0.48 per 100 words**

The study was conducted to determine the growth and yield performance of eggplant applied with Grower's Bet 1000+ Foliar Fertilizer, Pampanga, From November to April 2024. The experimental area was laid out using Randomized Complete Block Design (RCBD) there were three replications. The treatment are as follows: T0-Control; T1-Soil analysis; T2-250ml per 16L Foliar Fertilizer; T3-500ml per 16L Foliar Fertilizer and T4 - 750ml per 16L Foliar Fertilizer. This study aimed to investigate the impact of different Grower's Bet 1000+ Foliar Fertilizer treatments on the growth and yield of eggplants. The findings revealed that the application of Foliar Fertilizer did not significantly affect the timing of flowering in eggplants, as all treatments, including the control, exhibited the same average number of days until flowering. However, when it came to plant height at harvest, the Grower's Bet 1000+ Foliar Fertilizer (T2) 250 ml per 16L as the most effective treatment, resulting in the tallest average plant height compared to the control and other treatments. This notable increase in plant height can (EC) be attributed to the nutrient-rich content of GB 1000+ FF, which enhances soil conditions and facilitates better nutrient availability. In GB 1000+ FF contributed to vigorous vegetative growth and ultimately led to taller eggplant plants.

**Acs8 · 164 words · 0 hedges · 0.0 per 100 words**

The study on the performance of eggplant applied with different methods of Kuhol Amino Acid Application was conducted at [REDACTED], from December 2024 to March 2024 . The Mucho F1 eggplant variety was used as an experimental crop in the study. The experiment was laid out following Randomized Complete Block Design (RCBD) with three replications and four treatments, namely: T1 – Negative Control T2 – Foliar Application of Kuhol Amino Acid; T3 – Drenching Application of Kuhol Amino Acid; T4 – Inorganic Fertilizer (RR). Based on the results, there was no significant difference among the different methods of kuhol amino acid app in terms of plant height, weight of marketable and weight of non-marketable fruit, number of marketable and non-marketable fruits, and computed yield. Different methods of kuhol amino acid application influenced [G] the plant height, weight of fruits, number of marketable and non-marketable fruits, and computed yield. The researcher ~~recommends~~ (FP-Rec) foliar and drenching application of kuhol amino acid. because, numerically, it consistently produced the highest yield performance.

**Acs9 · 170 words · 2 hedges · 1.18 per 100 words**

A field experiment was conducted from December 2024 to January 2025 in [REDACTED], to evaluate the effects of varying soaking durations in Azospirillum spp. biofertilizer on the early growth and development of sugarcane (Saccharum officinarum) setts. The study employed a Randomized Complete Block Design (RCBD) with four treatments: T1 (24-hour water soak, control), T2 (24-hour Azospirillum spp. soak), T3 (20-hour Azospirillum spp. soak), and T4 (16-hour Azospirillum spp. soak), each replicated three times. Parameters assessed included percentage of survival, root elongation rate, root

area, fresh root and shoot biomass, and root-to-shoot ratio. Results indicated that the 24-hour Azospirillum spp. treatment (T2) significantly enhanced survival rates on Days 8 and 11, root elongation rate, root area, fresh root and shoot biomass, compared to the control. However, no significant differences were observed in root-to-shoot ratio among the treatments. These findings suggest (EC) that soaking sugarcane setts in Azospirillum spp. biofertilizer for 24 hours can (EC) effectively promote early growth by enhancing root development and biomass accumulation, thereby supporting improved propagation success.

**Acs10 · 246 words · 0 hedges · 0.0 per 100 words**

The field study was [G] conducted from January 2023 to April 2023 at [REDACTED] to evaluate the performance of rice as supplemented with foliar fertilizer. The experimental design used is a randomized complete block design (RCBD) with three treatments replicated three times. The following treatments were used: T1: recommended rate of NPK based on soil analysis; T2: recommended rate of NPK + 19-19-19; and T3: recommended rate of NPK + 14-14-14. The results of the study revealed the following: a significant difference was observed in the number of productive tillers, actual yield per sample area, and computed yield per ton/ha. No significant differences were observed in terms of the number of days from transplanting to flowering, plant height, number of unproductive tillers, length of panicle, number of filled grains, number of unfilled grains, and weight of 1,000 seeds. In conclusion, T3: NPK+14-14-14 was the most compelling treatment among all treatments in terms of producing the highest number of productive tillers and obtaining the heaviest yield per ton/ha. Cost and return analysis revealed that T3: NPK+14-

14-14 had the second lowest cost of production but had the highest net income, with a corresponding amount of Php 66,089 and an ROI of 97.53%. On the other hand, insignificant differences were noted in terms of the number of days from transplanting to flowering, plant height, number of unproductive tillers, length of panicle, number of filled grains, number of unfilled grains, and weight of 1,000 seeds.

## Agriculture, Animal Science (Aas1–Aas10)

### Aas1 · 203 words · 0 hedges · 0.0 per 100 words

The field was [G] conducted for 35 days from April 6, to May 10, 2023 in [REDACTED], [REDACTED] to determine the effect of feeding para grass mixed with varying levels of Gmelina arborea on the growth performance of rabbits. The experimental design used was a randomized complete block design (RCBD) with three treatments replicated three times. The following treatments were used: T1 ([Control] 60% Para grass + 40% Concentrate Feeds), T2 (20% GA Leaves + 40% Para grass + 40% Concentrate Feeds), and T3 (30% GA Leaves + 30% Para grass + 40% Concentrate Feeds). The results of the study revealed the following: No significant differences were observed in terms of mean initial weight (kg), mean final weight (kg), mean gain in weight (kg), mean DM intake (kg), or feed conversion ratio. The student researcher concluded that T2 (20% GA Leaves + 40% Paragrass + 40% Concentrate Feeds) was the most compelling treatment among all treatments for the rabbits in terms of outperforming the data numerically. The 20% Gmelina arborea + 40% Para grass + 40% Concentrate Feeds greatly contributed to performances in terms of

obtaining the highest mean final weight, mean gain in weight, and highest net income.

### Aas2 · 182 words · 0 hedges · 0.0 per 100 words

To evaluate the effectiveness of treating Pekin ducks with fermented okara extract (Glycine max) and track the growth of the ducks during the study. From first day up to last day of the study was carried out at the Poultry Project of Pampanga State Agricultural University (PSAU), located in Magalang, Pampanga. Ninety heads of week-old Pekin Duck were used in the study in a single factor experiment arranged in Complete Randomized Design (CRD). There were (3) three treatments, replicated three (3) times with ten (10) birds per replication. The researcher performed a one-way analysis of variance (ANOVA) on the gathered data. The average daily gain, feed conversion ratio, mean weight gain, mean feed consumption, and mean water intake all influenced the overall growth performance, according to the results. Fermented okara extract was ingested through drinking water. But when it came to mean ultimate weight in the treated birds, the fermented okara extract had [C] the biggest impact, resulting in the heaviest weight. Additionally, the total return result showed [G] a promising future for fermented okara extract at a level of 60 ml (166.00 PHP).

### Aas3 · 296 words · 2 hedges · 0.68 per 100 words

A sixty three (63) days study was conducted to determine the growth performance of mallard duck fed with dietary mixed ration in [REDACTED]. A total of ninety (90), 21 days old male mallard duck were used in the study. A single factor experiment in Randomized Complete Block Design (RCBD) was

used in the study. Mallard ducks were randomly divided into three (3) treatments having ten (10) mallard ducks per treatment with three (3) replications. Treatment 1 - commercial feeds; Treatment 2 - 50% commercial feeds and 50% rice bran mixed with water spinach; and Treatment 3 - 50% commercial feeds and 50% rice bran mixed with golden apple snail. Data were analyzed by using Analyzed of Variance in RCBD through the Statistical Tool for Agricultural Research (STAR) application. Treatments were compared using the Least Significant Difference (LSD). The result of the study showed in the treatments that there were no significant differences observed in feed consumption, which implies (EC) that treatments were comparable on parameters. However, final weight, gain in weight, average daily gain in weight and feed conversion ratio had highly significance differences at ($P<0.01$). This indicates that ducks fed with 100% commercial feeds rice bran mixed with golden apple snail performed better in final weight, gain in weight, average daily gain in weight and feed conversion ratio. In conclusion, proved the treatment 1(100% CF) are the most efficient compared to treatment 2 where 50% commercial feeds and 50% rice bran mixed with water spinach and treatment 3 where 50% commercial feeds and 50% rice bran mixed with golden apple snail. It can (EC) also be concluded that D1 rice bran mixed with water spinach and with golden apple snail used in the study ~~can~~ (FP-Cap) help minimize the cost of production, specifically the feed cost.

**Aas4 · 161 words · 1 hedges · 0.62 per 100 words**

The study was conducted for 30 days, from April 26, 2023, to May 25, 2023, at [REDACTED]. The study aimed to determine the average initial weight, average final weight, average gain in weight, average daily gain (ADG), average feed

consumption, average feed conversion ratio (FCR), average water consumption, and cost and return of the study. The study used eighteen (18) F1 New Zealand rabbits at 2 months of age. A Completely Randomized Design (CRD) was used in the study. The rabbits were divided into 3 treatments, 2 replications, and 6 rabbits per treatment. The experimental treatments were as follows: T1 = Pure Water (Control); T2 = 5 grams of ginger powder/L of water; and T3 = 8 grams of ginger powder/L of water, which were replicated three (3) times. The statistical analysis suggests (EC) that there is no significant difference among the treatments with regard to their initial weight, gain in weight, average daily gain, and feed conversion ratio of the experimental rabbits.

**Aas5 · 316 words · 2 hedges · 0.63 per 100 words**

Protein is the most costly and restrictive ingredient in feed formulations, and the price of traditional sources of protein has risen significantly due to yield factors and competition between humans and animals. In order to meet the demand for highly nutritious animal food, the future ~~will~~ (FP-Gen) drive animal production systems to find new sources of high-quality and sustainable protein feed raw materials. Fortunately, insect farming can (EC) be seen as a potential (EC) solution for filling the protein gap for feed and food resources in a more sustainable manned compared to other animal protein sources. In particular, there is an increasing interest in the utilization of Black Soldier Fly (BSF) larvae. The black soldier fly (BSF; Hermetia illucens) has been studied for its capability to convert recycling food and animal waste into high quality protein animal feeds. A total of 4,500 grams of nine-day-old (3rd instar) Black Soldier Fly larvae used in 3 experimental group randomly. Three (3) treatments

replicated three times used for the study, namely T1- Chicken Manure (Control), T2- Rabbit Manure and T3- Goat Manure. Every replication is composed of 500g of BSF larvae. Regarding in the yield parameters of Black Soldier Fly larvae, there was a highly significant observed among all treatment used in the study. BSF larvae fed with rabbit manure had the highest mean among all parameters under yield namely final live weight, final dry weight, biomass gain and survival gain. There is highly significant differences under nutritive analysis of Black Soldier Fly larvae. Among the results discussed, Crude Protein, Crude Fat, Ash Content and Metabolizable Energy has a significant results at 1% level. In Calcium, Phosphorus and Moisture Content had a comparable results in all treatment used in the study. The BSFL in treatment 3 (14.63% Crude Fiber) and treatment 2 (13.53% Crude Fiber) had a significantly higher crude fiber than the treatment 1 group which obtain 8.73% Crude Fiber.

**Aas6 · 209 words · 0 hedges · 0.0 per 100 words**

The study examines the germination and growth rate of pigeon pea (Cajanus cajan) under different scarification method for 30 days. The effects of various scarification method, including water in different temperatures, mechanical scarification (sand paper method) and a chemical scarification (sulfuric acid), were assessed through multiple parameters. The study analyzed the germination rate, days to germinate, plant height, length of roots, age to bear leaves, length of leaves, number of leaves, number of branches, mortality, and cost production analysis. The type of scarification method that showed prime results was [G] Treatment 3 (sand paper method). It was found to have a significant effect on the growth and germination rate of pigeon pea in terms of

days to germinate, plant height, a number of leaves, and length of leaves. Sulfuric was second in terms of the scarification method effectiveness for pigeon pea, The study treatments overall proclaimed a not significant results based from the analyzed data from each treatments except from these group of data such as from the length of leaves from week 2 and week 3 and the length of roots that are all expressed a highly significant results at 1% level and also the plant height from week 3 that also expressed a significant result at 5% level.

**Aas7 · 202 words · 3 hedges · 1.49 per 100 words**

This study was conducted over a 35-day period from May 10, 2023, to June 14, 2023, at [REDACTED]. The present study aimed to determine the growth response of crossbred New Zealand rabbits as influenced by varying levels of fermented Madre De Agua leaves. Eighteen (18), 2 months of crossbreed New Zealand rabbits were used in the study which were divided into three (3) treatments: T1 = 70% concentrate + 30% paragrass, T2 = 70% concentrate + 20% paragrass + 10% fermented Madre De Agua, T3 = 60% concentrate + 20% paragrass + 20% fermented Madre De Agua. A Completely Randomized Design was employed, and data was analyzed through ANOVA by Statistical Tool for Agricultural Research (STAR). Statistical analysis showed that there was no significant difference but showed numerical trends among the treatments in relation to initial weight, final weight, gain in weight, average daily gain, dry matter intake, and feed conversion ratio of the experimental rabbits In conclusion, all treatments are comparable in terms of growth, implying (EC) that Fermented Madre de Agua is a viable feed component under these conditions. Feeding 20% fermented Madre De Agua may (EC) improve final

weight, while feeding 10% fermented Madre De Agua may (EC) reduce dry matter intake.

### Aas8 · 186 words · 0 hedges · 0.0 per 100 words

A study was conducted to evaluate the survivability of mallard ducks (Anas platyrhynchos) through growth performance assessment fed with different ration composition at brooding stage from day 1 to day 21 of age. T1 served as the control treatment fed with chick booster while ducks fed with T2 were given 75% rice porridge and 25% fish entrails and for T3 with 50% rice porridge and 50% fish entrails. The results revealed that treatment 2 consistently demonstrated the highest mean gain in weight, mean final weight, average daily gain, and feed consumption during the initial 14-day period and days 15 to 21. However, T1 exhibited the lowest feed conversion ratio (FCR) and mortality rate, ~~indicating~~ (FP-Gen) superior feed efficiency and improved survivability to T2. Based on the results of the study, it is ~~recommended~~ (FP-Rec) to feed mallard ducks with 75% rice porridge and 25% fish entrails during brooding stage at day 1 to day 21 for enhancing weight gain and achieving higher final weights in mallard ducks. Further research ~~can~~ (FP-Gen) explore the long-term effects and cost-effectiveness of these treatments and their impact on meat quality and reproductive performance.

### Aas9 · 188 words · 0 hedges · 0.0 per 100 words

The study was conducted at [REDACTED] from April 25 to April 30, 2023 to find out the [TITLE REDACTED] where a total of nine hundred maggots (900) were used. The experiment was laid out using Complete Randomized Design (CRD) with three (3) treatments, namely, T1 –Carbonated rice hull and saw dust, T2 –

Diatomaceous Earth, and T3 –Neem tree extract. The nine hundred maggots were divided into three (3) treatments at one hundred maggots (100) per treatment each treatment was replicated three (3) times with one hundred maggots (100) per replication. The results disclosed that the percentage of mortality and moisture content were statistically significant at 1% (P<0.01) and (P<0.05) level with 48 hours after the post-treatment. However, mortality rate and moisture content were not significant after 6 hours and 24 hours. Therefore, it is ~~recommended~~ (FP-Rec) for future researchers to add Diatomaceous earth and neem tree extract and check the humidity. Extend data collection for 72 hours and 96 hours for future study.

**Aas10 · 283 words · 2 hedges · 0.71 per 100 words**

The study was conducted at [REDACTED], for 35 days, from April 16 to May 20, 2023. The study was carried out to search for alternatives to protein sources. The objective of the study was to determine the growth response of F1 New Zealand white rabbits as influenced by the roughage-to-concentrate ratio. A total of eighteen (18) heads of six (6)-week-old female New Zealand White Rabbits were selected and distributed into three (3) groups, each treatment replicated thrice with 2 rabbits per replication. Results showed no significant difference (P<0.05) in final weight, gain in weight, and average daily gain in all the treatment groups. The control group (T1) was fed 70% guinea grass and 30% concentrates. For T2 and T3, the guinea grass was replaced with fresh azolla on an equi-weight basis at levels of 10 and 20 percent, respectively. In conclusion, based on the results, it can (EC) be stated that the ratio of 70% guinea grass (GG) + 30% concentrate (C), the control group, had a significantly higher (P<0.01) feed conversion ratio, while rabbits in treatment T3 (50%

guinea grass, 20% azolla, and 30% concentrate) had a significantly higher (P<0.01) dry matter intake. The researchers still ~~recommended~~ (FP-Rec) incorporating 50% Guinea grass + 20% Azolla + 30% concentrates for rabbit growth performance, considering the results show that this treatment performs well and doesn't differ from treatment 1, which is the farmer practice. The researchers also saw that the treatment of 50% Guinea grass + 20% Azolla + 30% concentrates has potential (EC) as an alternative feed ration for rabbits without adverse effects, and considering that azolla ~~can~~ (FP-Cap) help breeders to lessen the feed cost in production.